L-7 Radar Design Project
Spring 2026

Specifications – Draft Rev-A

2/18/26

EEE5557 Intro to Radar Systems

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- You must form design teams (one or two persons) and submit your proposed design to be evaluated by a technical evaluation board comprised of Dr Jones and his senior grad RA’s
- Register your team members with DrJ and receive a group#
  - Proposals will not be accepted from unregistered groups

Requirements

- This document presents the design requirements, goals and constraints
- **All radar designs must be fully compliant**
  - **All specification requirements must be met**
  - Where **goals** are given, these do not have to be met; but the customer gives favorable consideration if the goal is met
    - All goals are equally weighted and proportional credit will be given for partial achievement of goals
      - e.g., performance = 50% of the goal, receives 50% credit for achieving goal

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      - e.g., performance = 50% of the goal, receives 50% credit for achieving goal
  - **Failure to submit a compliant design will result in disqualification**
    - A maximum grade of 75% is possible for non-compliant designs
      - Grade to be based upon the quality of the trade studies performed

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    - 1st place = 5 pts; 2nd place = 4 pts; and 3rd place = 3 pts
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- **Winners must present their designs during a 20 min ppt presentation on April 27th (either in present or by Zoom for off-campus students only)**

Power Point Presentation

- Your proposal must be documented in a power point presentation
  - You must present your radar design in a series of bullet and graphics charts
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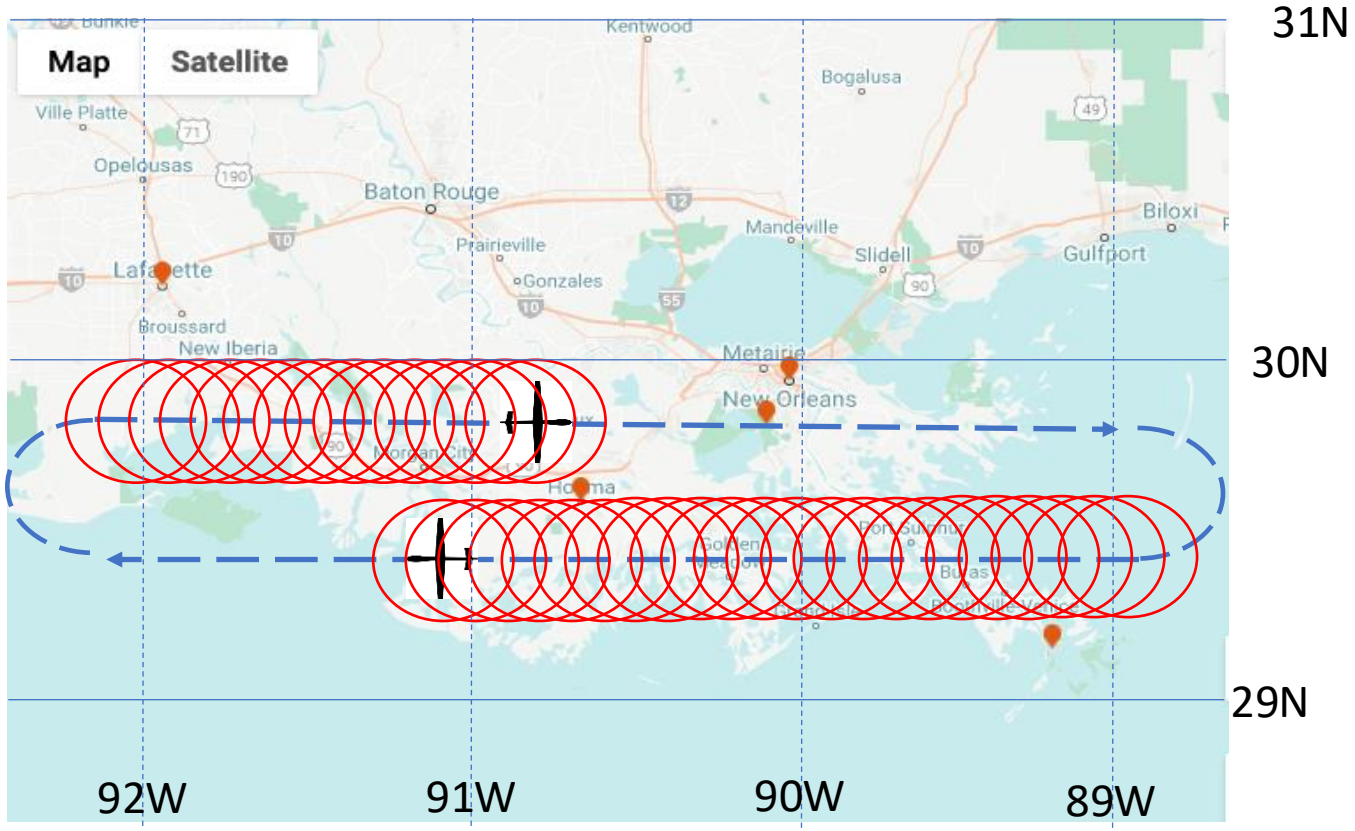
- **Final power point presentations are due Monday 4/2th @ noon (no exceptions)**

Specification Details

- The following specification provides the design inputs and system requirements
  - Where parameters are marked as “???” this means that it’s the designer’s choice – based upon a system optimization trade study
- This specification document is **<u>Draft Rev-A (2/18/26)</u>**

Project Scenario

Two predator drone unmanned aerial vehicles (UAVs) will fly a “race-track” trajectory while providing 24-hour radar surveillance to detect smugglers operating in the wetlands of the Louisiana Bayou

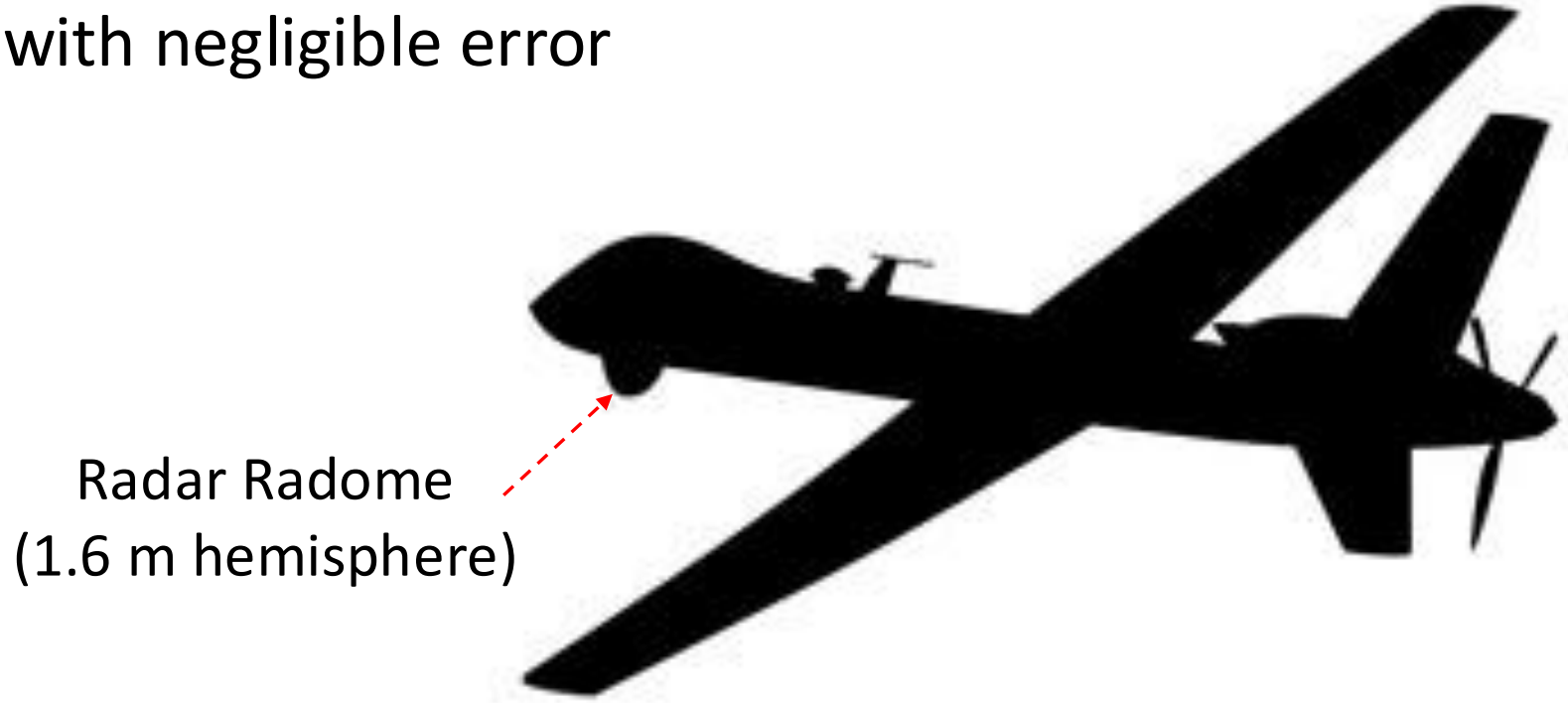


Predator UAV

MQ-18 Predator Drone

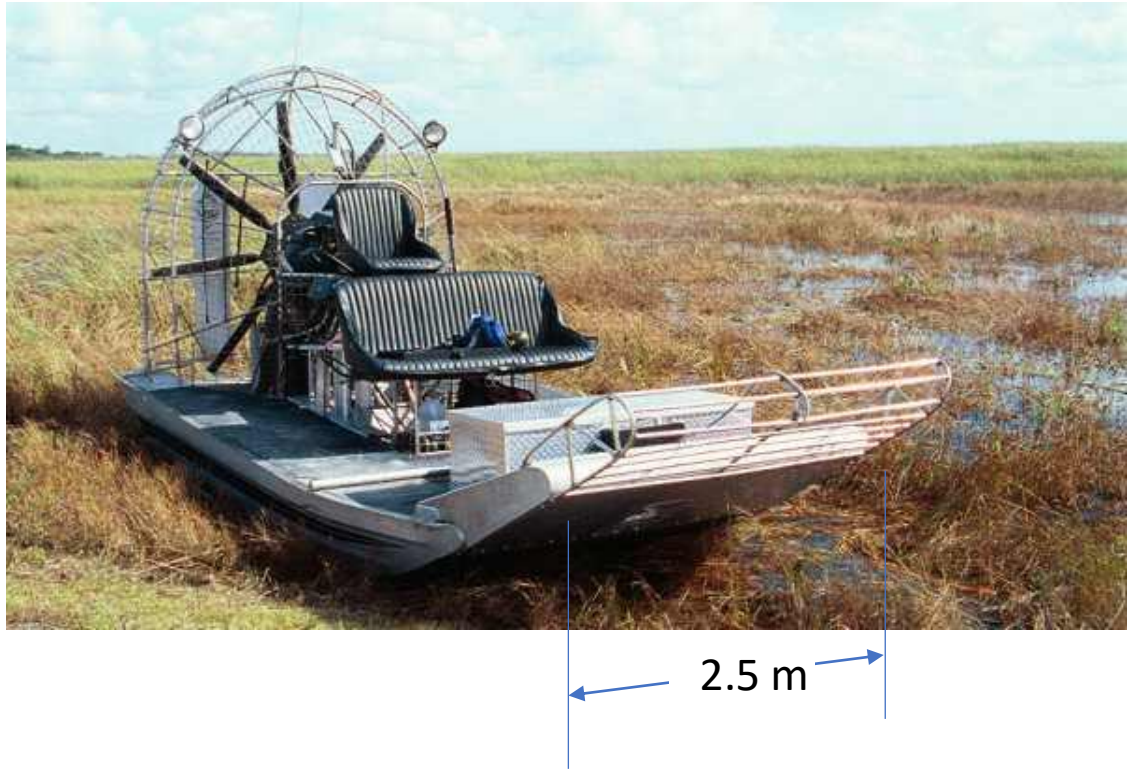
Payload: 450 pounds (204 kilograms)
Speed: cruise speed around 84 mph (70 knots), up to 135 mph
Range: 770 miles (675 nautical miles)
Ceiling: 25,000 feet (7,620 meters)

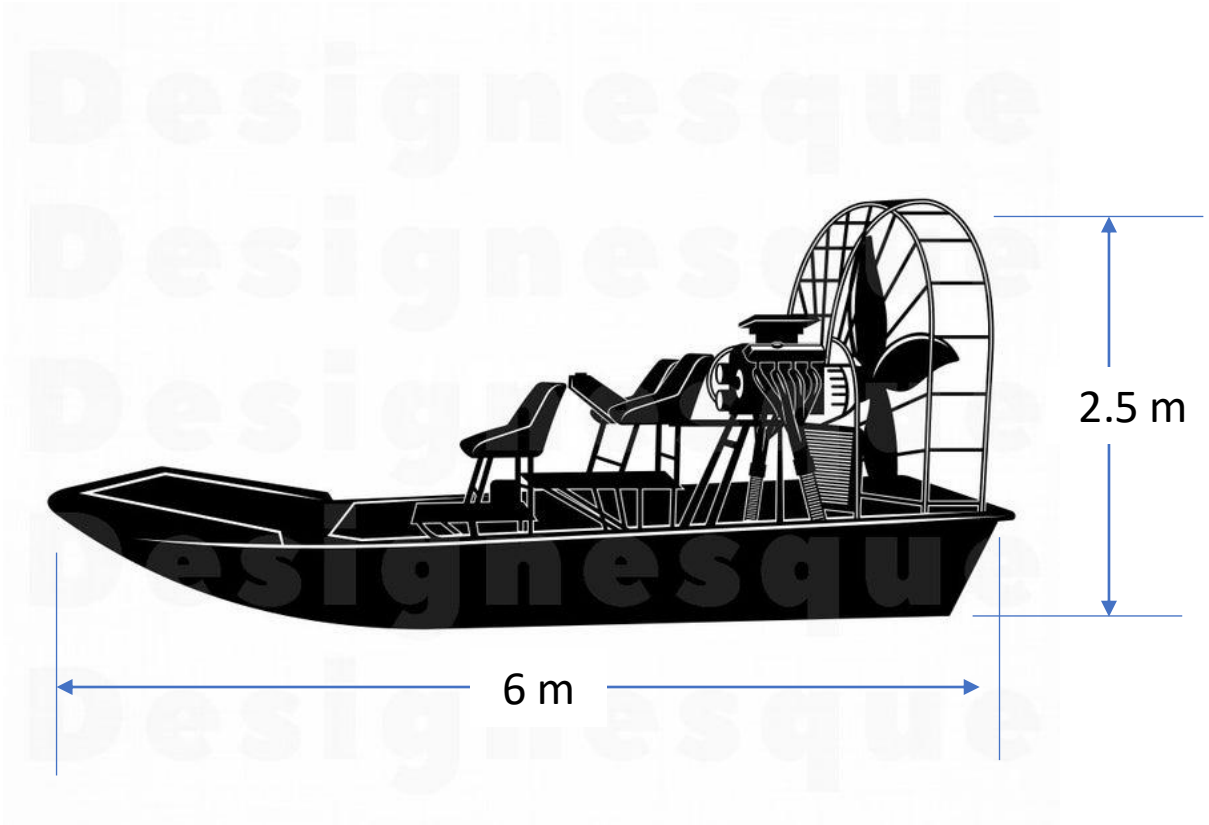
- **Predator UAV Operations:**
- **UAV: altitude**: 2.5 – 7.5 km (in steps of 0.5 km) ??? (straight and level flight);
- **ground speed**: 115 km/hr, (31.9 m/s)
- **Assume GPS navigation**: UAV location (Lat/Lng) and ground speed are known with negligible error



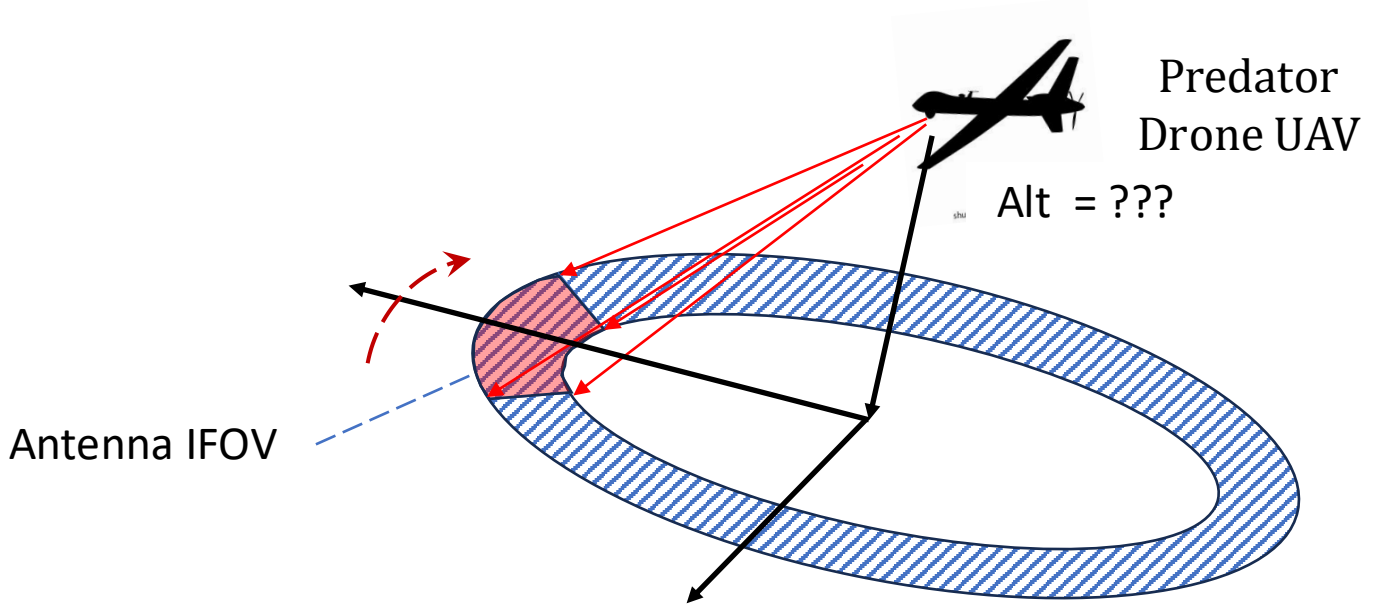
Radar Targets

- Airboats smuggling “contraband” into the United States from the Gulf of America at speeds of 50 to 170 km/hr, (87.4 m/s)

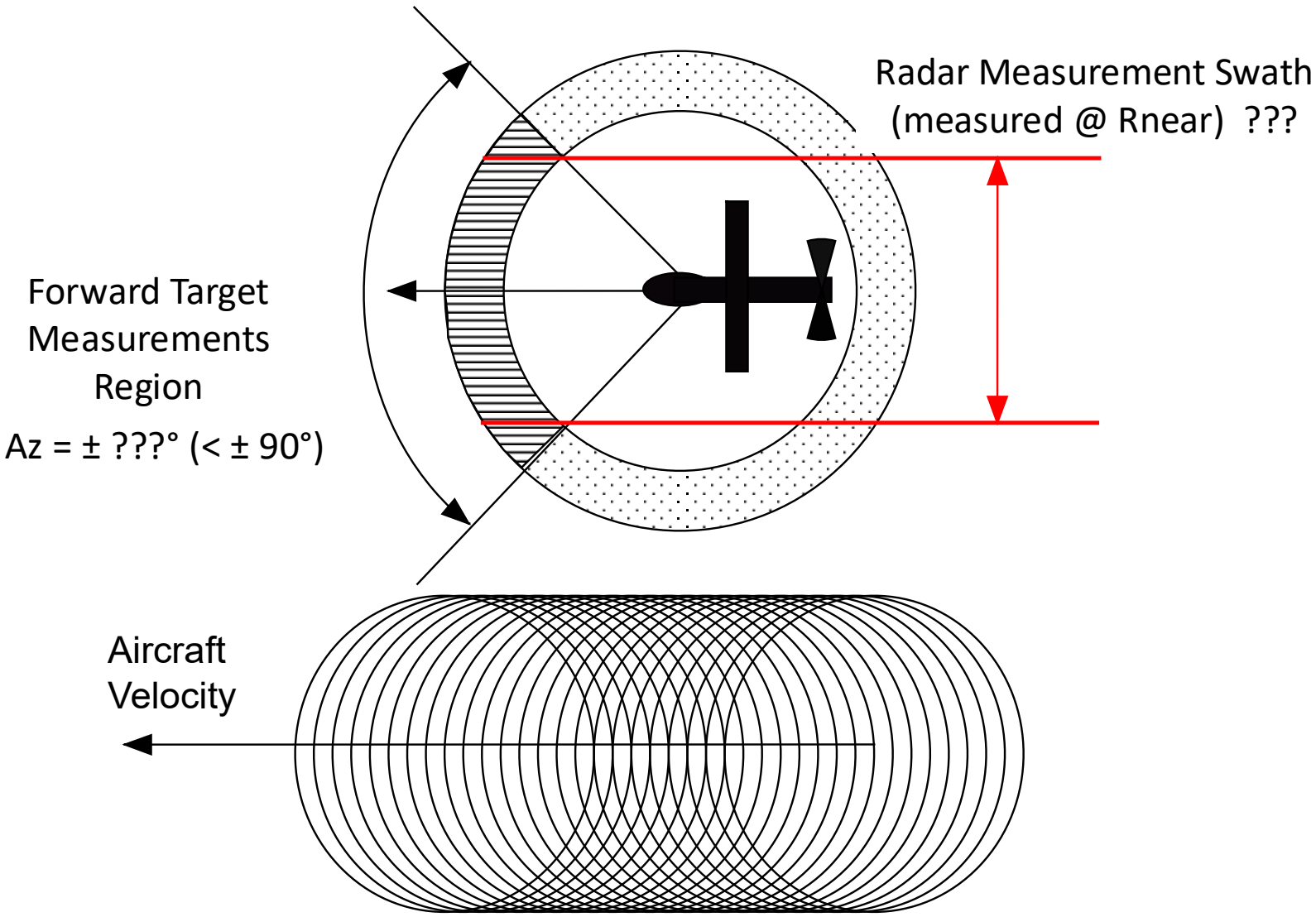




- **Airborne Pulse Radar Design**
- Design a conical-scanning radar operating on a “Predator drone" unmanned aerial vehicle (UAV) to provide surveillance of wetlands in the Louisiana Bayou to detect non-cooperative vehicle targets (air boats) smuggling contraband during night and low visibility conditions



Top View



Multiple-scan Pattern

Pulse Radar System Specs

- Radar measurements shall be made looking forward (or fore & aft ???)
  - Active radar measurements azimuth scan = ??? (< ±90° equal in both fore & aft directions)
  - Measurement Swath: ≥ 10 Km spec; 50 km goal

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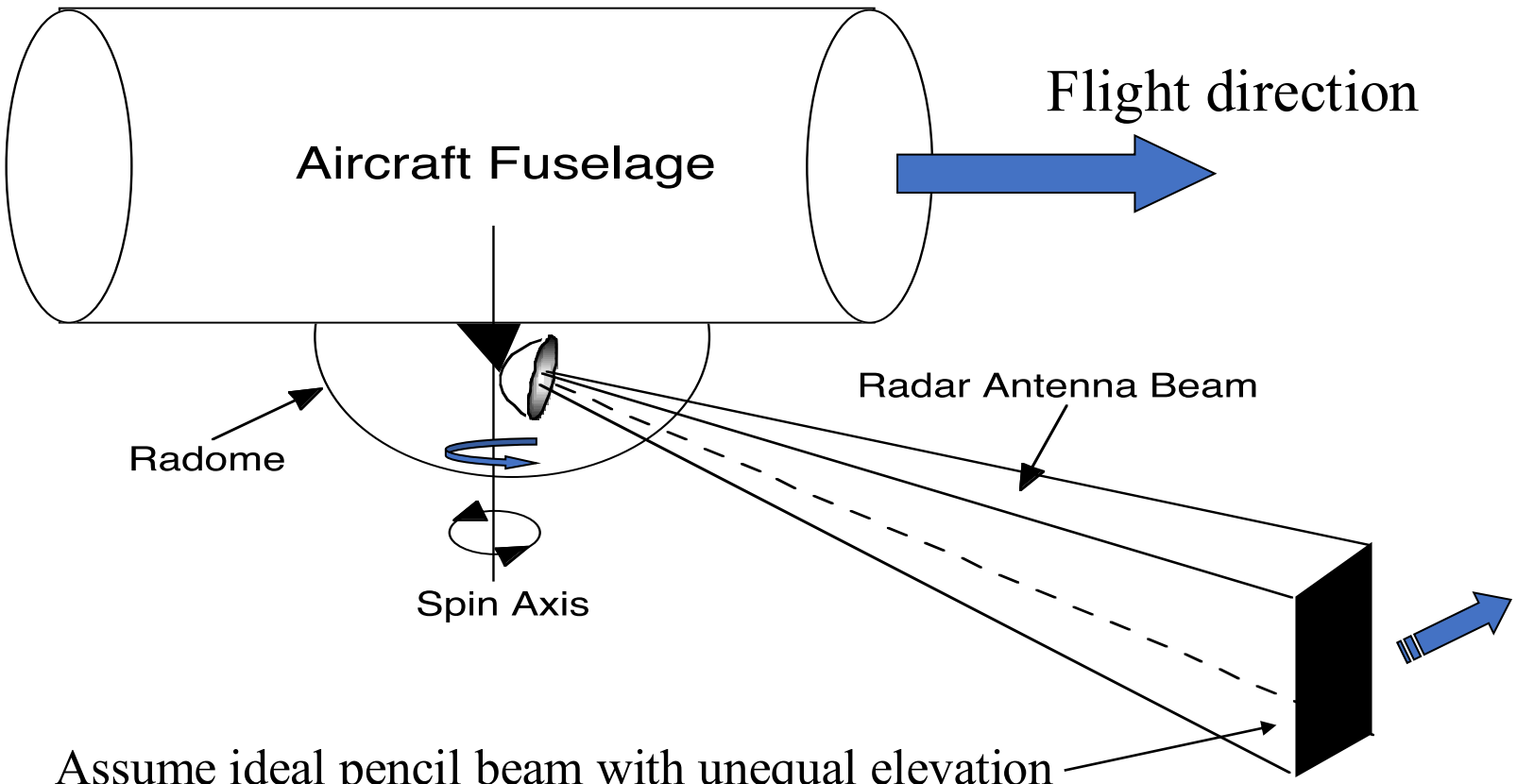
- Radar measurements shall be made looking forward (or fore & aft ???)
  - Active radar measurements azimuth scan = ??? (< ±90° equal in both fore & aft directions)
  - Measurement Swath: ≥ 10 Km spec; 50 km goal
- Transmitter:
  - Freq: ??? either: 37 GHz, 90 GHz or 150 GHz;
  - P_t ≤ 1,500 W (goal is ≤ 10 W), duty cycle constraint: ≤ 10 %;
  - PRF ≤ 500 kHz
  - Xmt pulse width (tau): ???

Pulse Radar System Specs

- Transmitter:
  - Freq: ??? either: 37 GHz, 90 GHz or 150 GHz;
  - $P_t \leq 1500$ W (goal is ≤ 10 W), duty cycle constraint: ≤ 10 %;
  - PRF ≤ 500 kHz
  - Xmt pulse width (tau): ???
- During target detection, <u>only one pulse on the surface at a time, but PRF may be range ambiguous</u>
- During Doppler mode, there are no restrictions on the # pulse simultaneously on surface; also, duty cycle constraint is relaxed to ≤ 20%
- Note that target detection and doppler may be simultaneous (parallel processing) or sequential (??? designer choice)

Airborne Conical Scanning (rotating) Radar Fan Beam Antenna



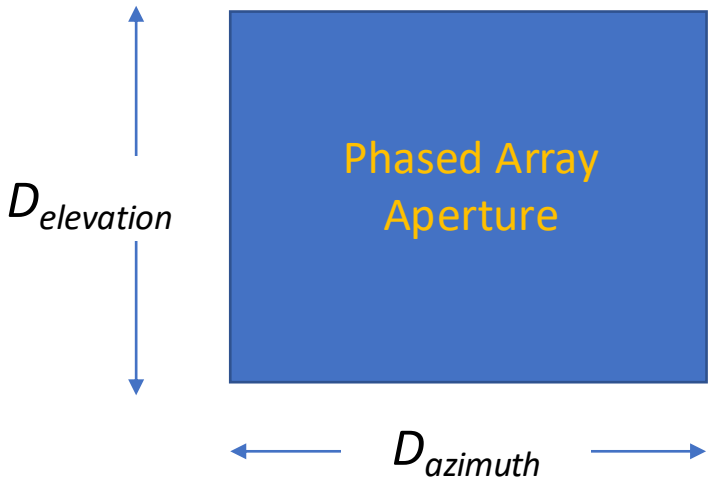
Assume ideal pencil beam with unequal elevation and azimuth HPBW's (??? designer's choice)

- Antenna aperture:

≤ 1.5 m ??? elevation by ≤ 1.5 m ??? azimuth

  - (constrained by size of hemispherical radome)
  - Goal = 0.1 m (but not necessarily both plane together)

- Aperture size defines antenna HPBW : β_{EL} ; β_{AZ}
  - $Beta_{ij} = 1.2*(lambda_j)/dia_i$, radians
    - Where *i*= El or Az and *j* = 37, 90 or 150 GHz



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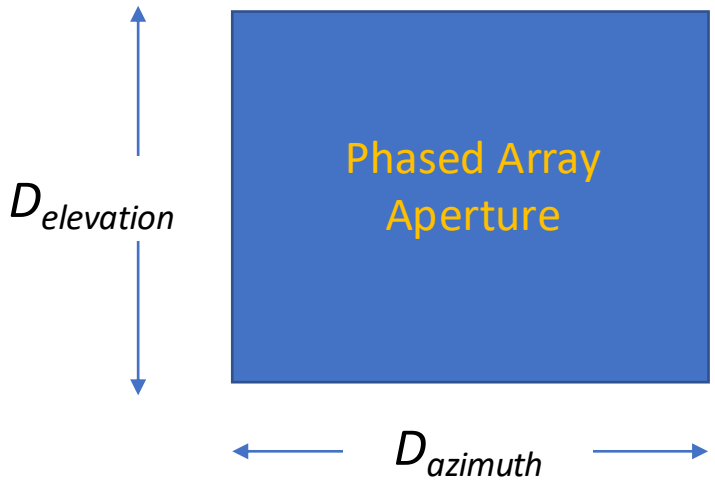
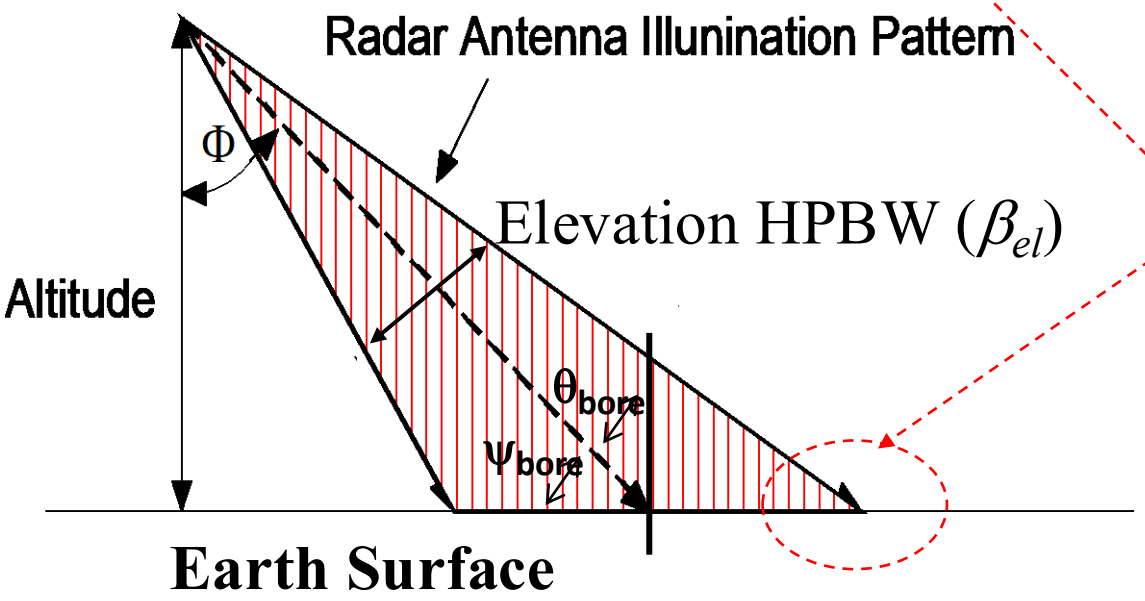
  - (may be elliptical beam: i.e., az & el HPBW may differ)



- Boresight: Elevation boresight grazing angle* ψ_{bore} = ???? ; but $\psi_{far} \geq 2°$



* Radar Grazing angle defined by the 2-way antenna pattern

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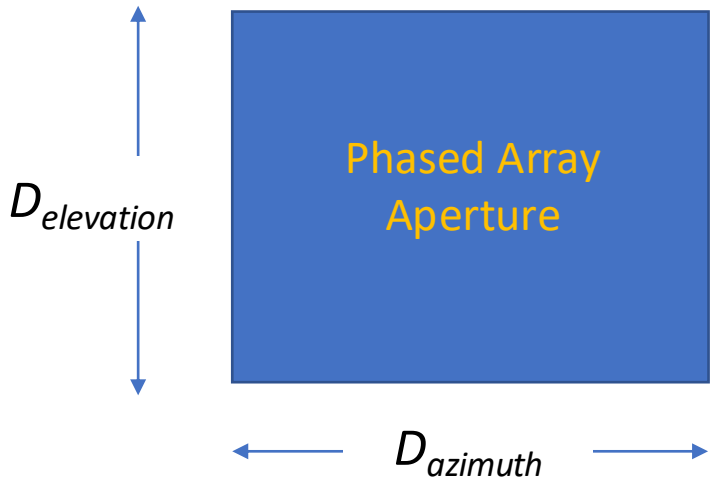
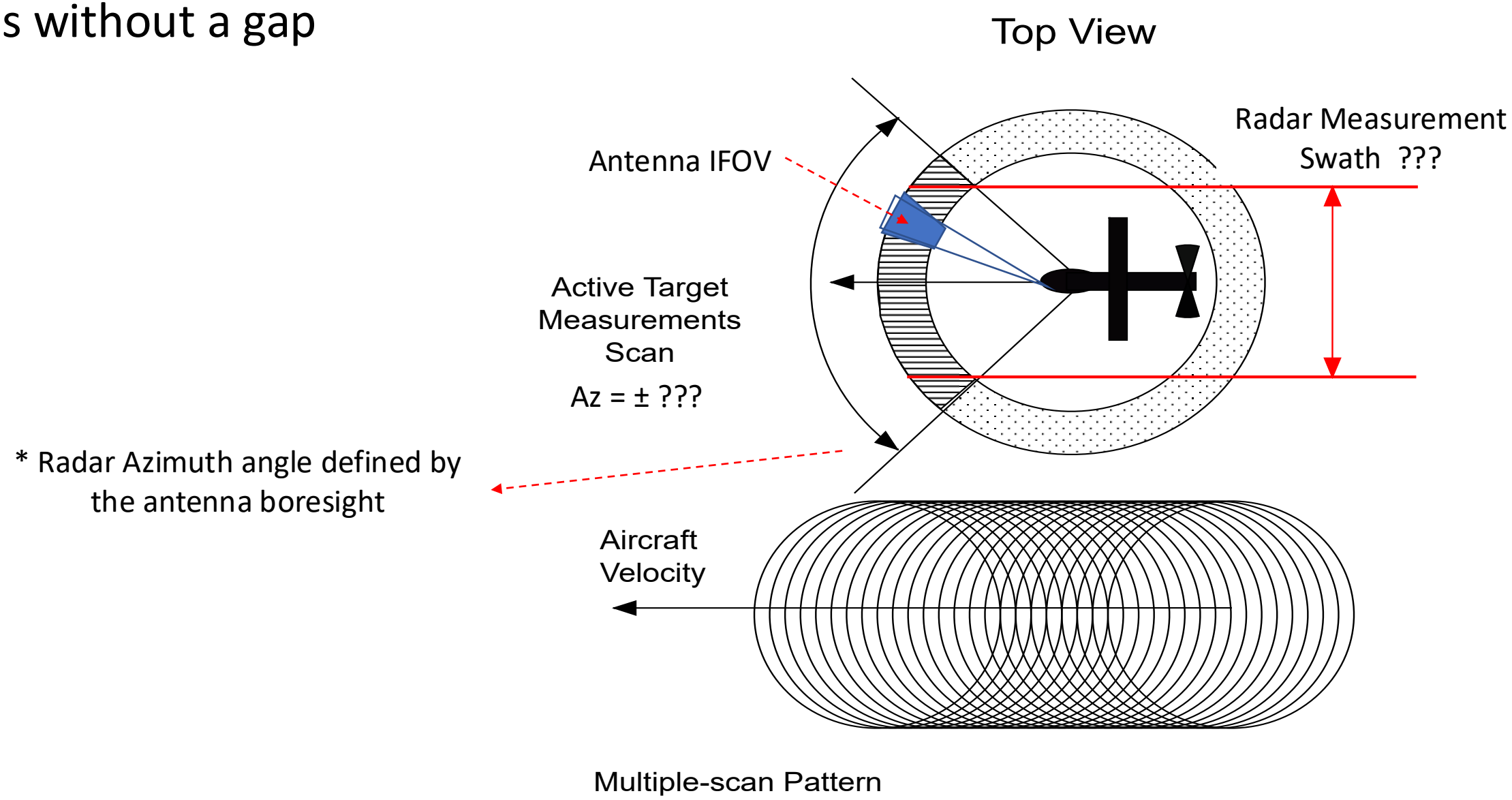


- Boresight: Elevation boresight grazing angle ψ_{bore} = ???? ; but $\psi_{far} \geq 2°$
- Antenna gain (one-way antenna pattern): = 0.6 * 4 pi / (Beta-el * Beta-az)
  - where Beta is the HPBW in radians
- Max. antenna scan rate ≤ 180 rpm
  - (hardware limit - may not exceed)

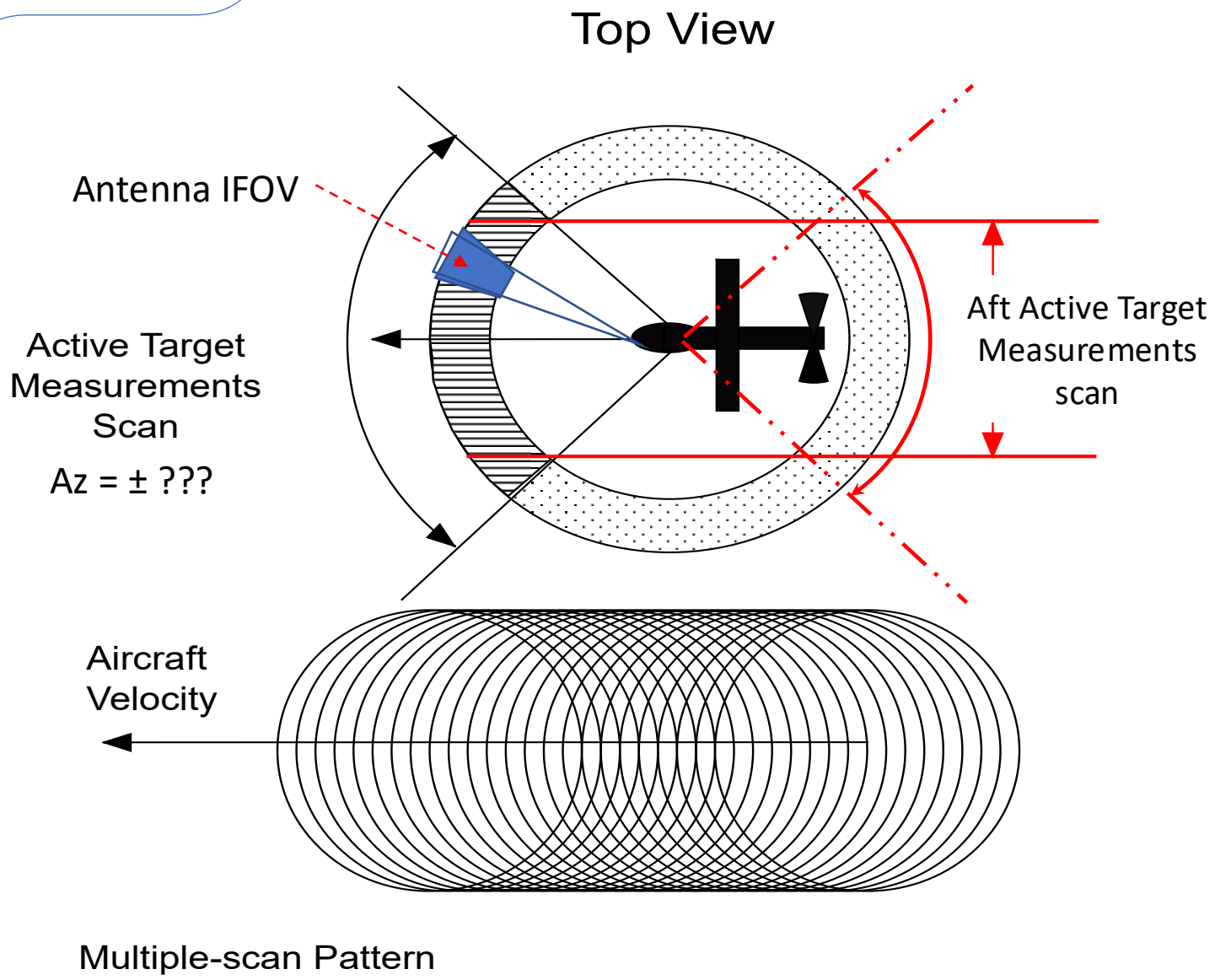
Antenna aperture efficiency

- Antenna Scan coverage: Along the active radar measurements swath looking forward (or fore & aft???), AZ < ±90° to provide contiguous range gates without a gap



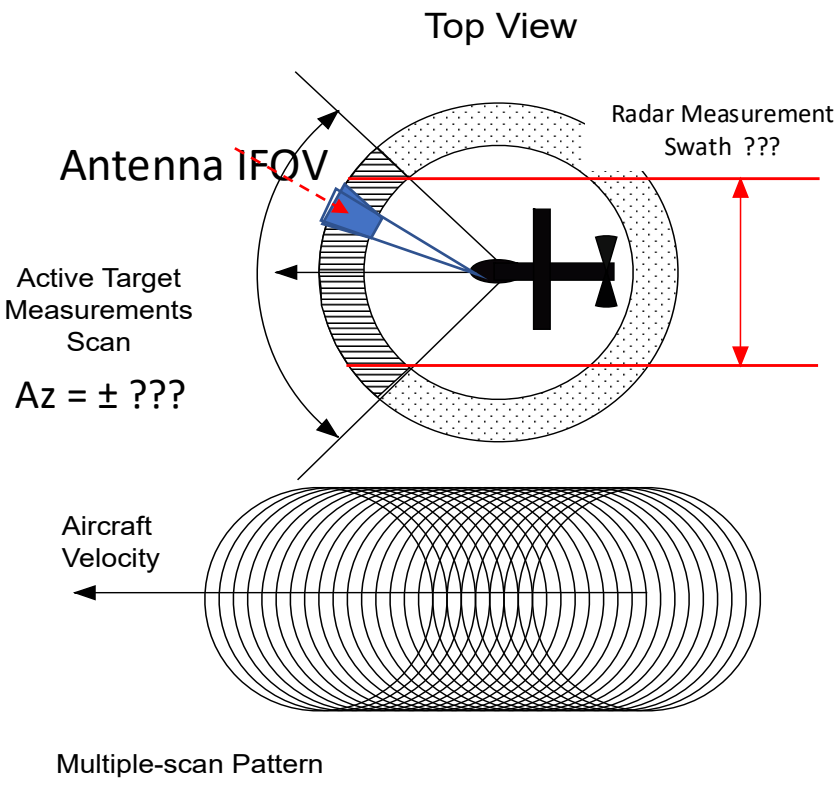
- Antenna Scan coverage: Along the active radar measurements swath looking forward (or fore & aft, AZ < ±90°) to provide contiguous range gates without a gap

The ± Az range is selected looking forward; and if both fore & aft operation are proposed, then ± Az must be the same.



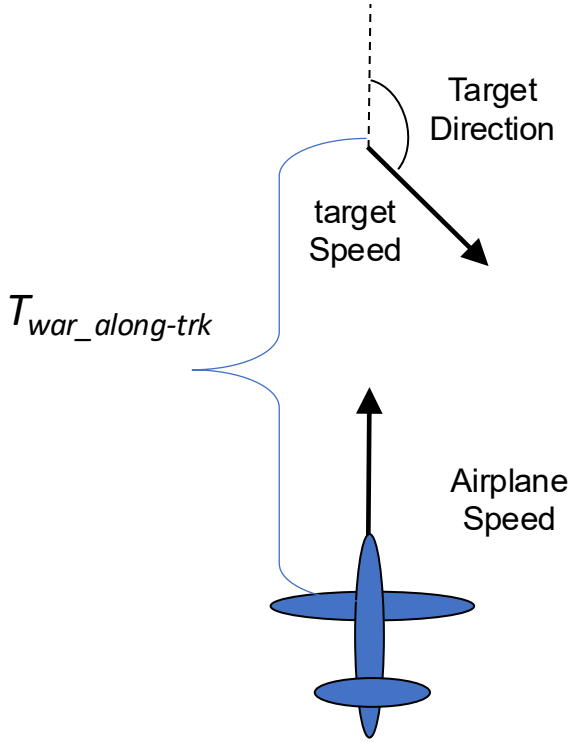
- Antenna Scan coverage: looking forward, (or fore & aft, AZ < ±90°) to provide contiguous range gates without a gap

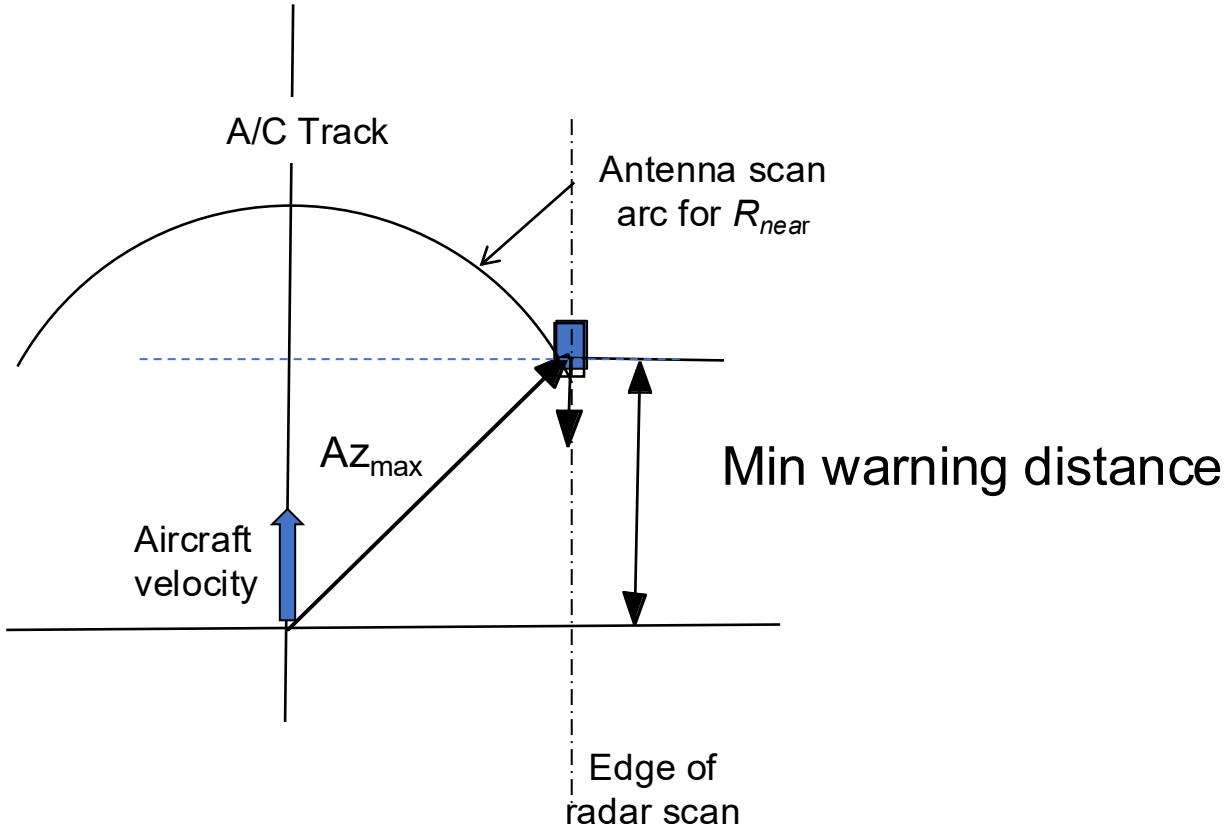
- Given that forward and aft looks are allowed, we need to identify which the requirements are impacted
- The P_d req'mt applies for only forward looks
  - i.e. , detection must occur for ≥ 3 scans (looking forward) while the target is illuminated by the radar
- Doppler measurements may <u>combine both forward & aft</u> scans to meet the Doppler resolution req'mts (TOT)



Target Warning Time Definition

Only, when looking forward, the radar Target detection "warning time" T_{WAR} specification applies

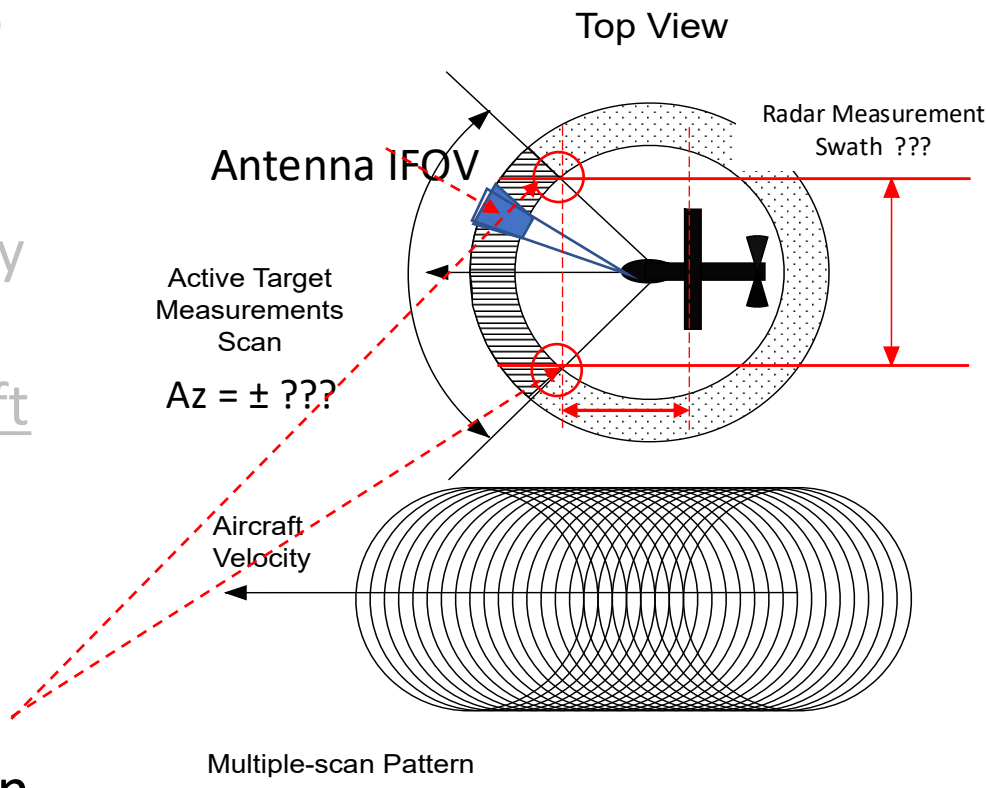




$$T_{war_min} = T_{war_along\text{-}trk} * \cos(Az_{max})$$

- Antenna Scan coverage: looking forward, (or fore & aft ???, < ±90° Az) to provide contiguous range gates without a gap

- Now that forward and aft looks are allowed, we need to identify how the requirements are impacted
- P_d req'mt applies for only forward looks
  - i.e. , detection must occur (looking forward) on every scan that the target is illuminated by the radar
- Doppler measurements may <u>combine both forward & aft</u> scans to meet the Doppler resolution req'mts (TOT)
- Target detection "warning time" T_{WAR} applies only to forward look
  - Note: worst-case condition occurs at edge of swath
- Detecting the target in ≥ 3 scans applies only to fore scan (and likewise the goal of 10 continuous scans applies only to the forward look)



- Antenna Scan coverage: looking forward, (or fore & aft, < ±90° az over 360° rotation) to provide contiguous range gates without a gap
- Overlap of antenna IFOV scan-to-scan will be required (fore-look) to assure that moving targets are not missed for ≥ 3 contiguous scans (goal = 10 scans)

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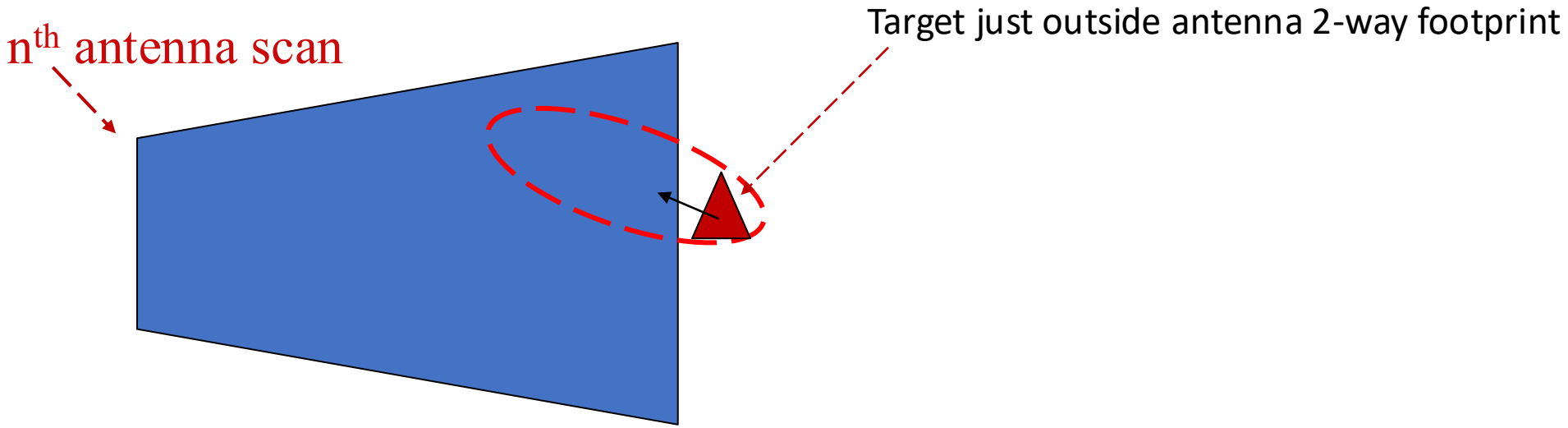
Example: Target not seen on n^{th} scans



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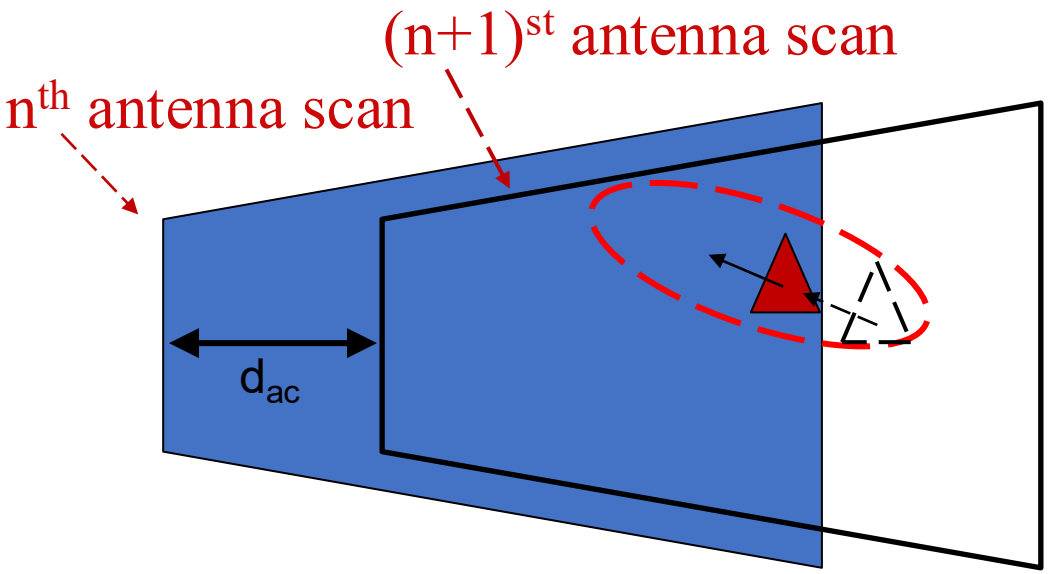
Example: Target seen: (n+1) scan



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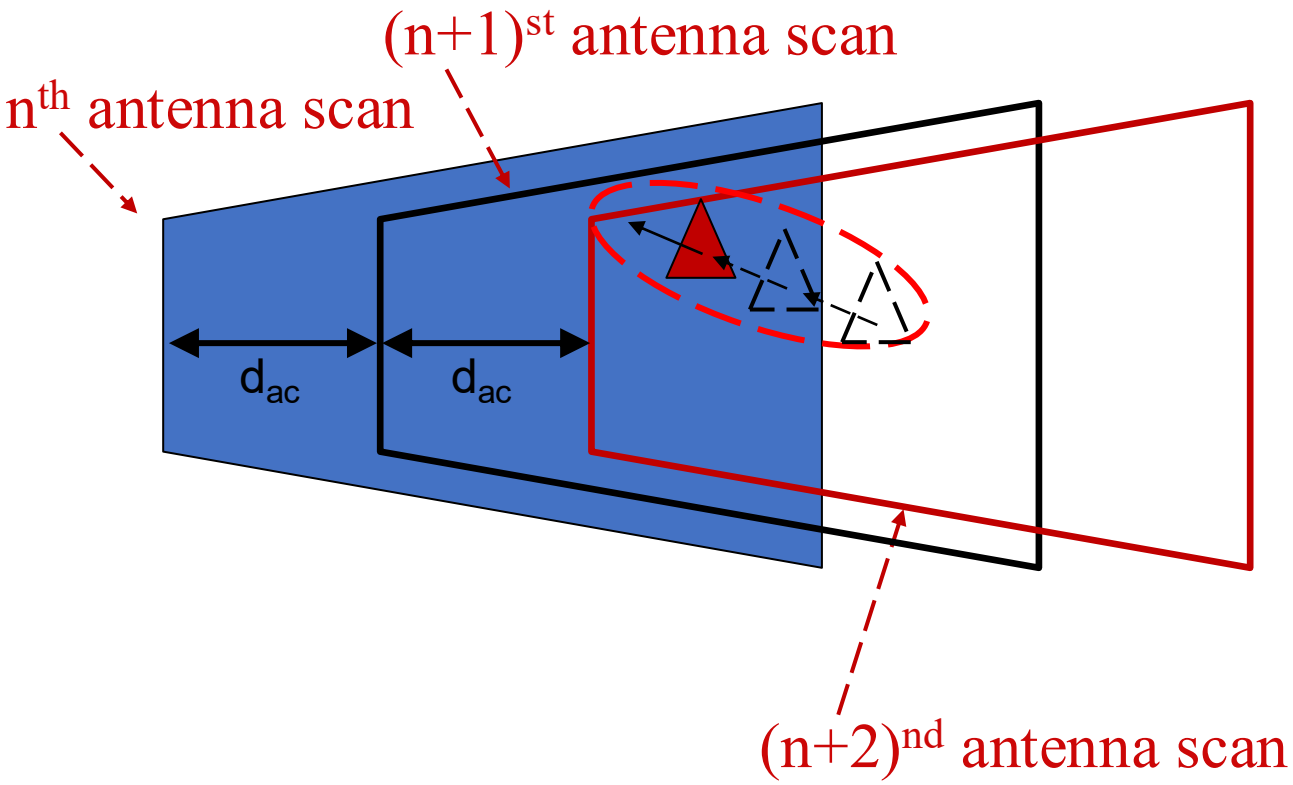
Example: Target seen on 2 scans: (n+1) & (n+2)



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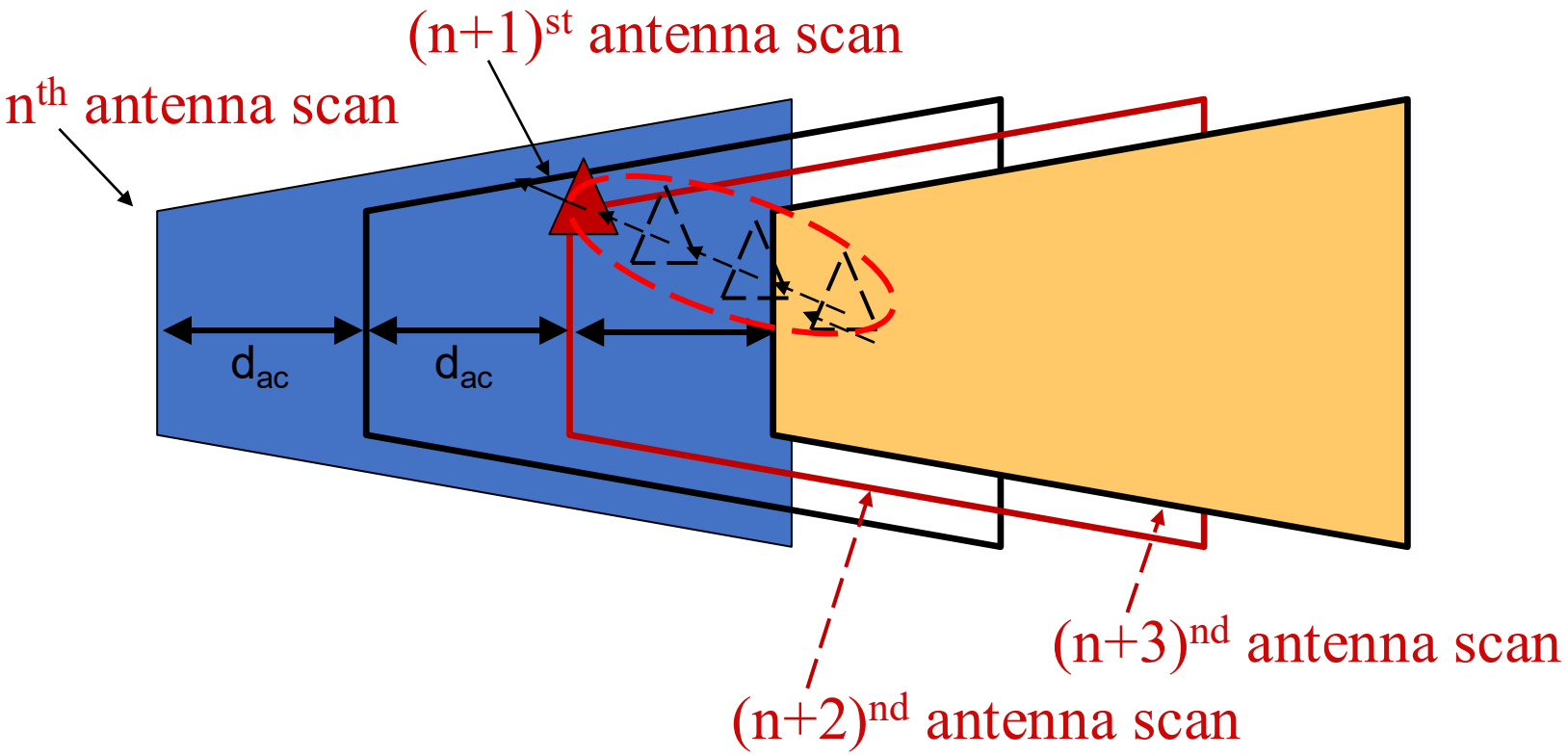

- Other System Specs:
- Receiver Bandwidth: pre-detection -3dB bandwidth B = 1/tau
- System Noise Temperature:

  T_{sys} : @ 37 GHz = 400 K; @ 90 GHz = 500 K; @ 150 GHz = 700 K
- Atmospheric propagation loss coeff:

  37 GHz: - 0.05 dB/km; 90 GHz: -0.15 dB/km; 150 GHz: -0.40 dB/km
  - Atmospheric loss only applies for Alt < 5 km
    - i.e., the max slant range for path loss = 5.0/sin(grazing_ang), km

Signal Processor: Simultaneous Range & Doppler Measurements

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  @ R_{near} , R_{bore} and R_{far}

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  - Target detection must be performed for cum target detection: P_d(2 of 3) ≥ 0.80 @ R_{near} , R_{bore} and R_{far}
  - Doppler measurement may combine pulses from multiple scans (and fore & aft looks) to meet spec
  - Avg. Time between False Alarms: ≥ [10 times the antenna scan period]
    - i.e., on average ≤ one false alarm/10-scans

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By using both fore and aft scans,

- The designer has the option to perform target detection only, while looking forward, with the radar operating in the pulse-radar mode
  - In this case, cum target detection (Pd(2 of 3) ≥ 0.80) must be performed first
  - After achieving Pd, radar may be switched to Doppler mode and Doppler measurements may be performed by combining pulses from multiple scans
  - **If this mode is proposed, then you must provide a timing diagram for one complete scan, which identified the time and duration of the radar mode changes (detection/Doppler)**

Radar Signal Processor, Sequential Range & Doppler Measurements

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- When looking aft, Doppler measurements could be performed, with the radar operating in the coherent high-PRF Doppler mode

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  - Doppler measurements may be performed by combining pulses from multiple scans
    - If this mode is proposed, then you must provide a timing diagram for one complete scan, which identified the time and duration of the radar mode changes (detection/Doppler)
  - When switching to Doppler mode, radar pars: Xmt pulse width and PRF may be changed to different values (than target detection mode) and multiple pulses may occur on surface simultaneously
    - However, RG ≥ max Target dimension for both modes

Radar Signal Processor, Sequential Range & Doppler Measurements

- If radar design cannot accommodate simultaneous ranging and Doppler, then time sequential target detection and Doppler measurements may be performed

- When looking aft, Doppler measurements could be performed, with the radar operating in the coherent high-PRF Doppler mode
  - Doppler measurements may be performed by combining pulses from multiple scans
    - If this mode is proposed, then you must provide a timing diagram for one complete scan, which identified the time and duration of the radar mode changes (detection/Doppler)
  - When switching to Doppler mode, radar pars: Xmt pulse width and PRF may be changed to different values (than target detection mode) and multiple pulses may occur on surface simultaneously
    - However, RG ≥ max Target dimension for both modes
  - The max PRF ≤ 500 kHz and the pulse-Doppler duty cycle ≤ 20%

More on Target detection:

- Signal processor uses post-detection (non-coherent) pulse integration of "pulses on target" for a single antenna scan and must use multiple scans to meet the P_d(2 of 3 scans) requirement:
- Also, the requirement is to detect the target in at least 3 consecutive scans to establish a target ground track, and the goal is to observe the target on 10 consecutive scans
  - Bonus evaluation points will be assigned for this enhanced capability

More on Target detection:

- Signal processor uses post-detection (non-coherent) pulse integration of "pulses on target" for a single antenna scan and may use multiple scans to meet the P_d(N of M scans) goal
- The requirement is to detect the target in at least 3 consecutive scans to establish a target ground track, and the goal is to observe the target on 10 consecutive scans
  - Bonus evaluation points will be assigned for this enhanced capability
- Probability of Detection: requirement ≥ 0.80 for cumulative P_d(2 of 3 scans) - regardless of location of target in antenna footprint i.e., near range to far range at any azimuth
- The {(target)-to-(clutter+noise) = T/(C+N) } power ratio must be used as the worst-case signal-to-noise (S/N) in all analyses
  - You must provide the "worst case" analysis to support your design @ near range, @ boresight and @ far range

- **Minimum Target Detection Time (T_{WAR}):** Using the <u>worst-case target velocity (speed and direction of travel) and swath location</u>, the **requirement** is to detect the target **at least 6.0 sec** before the aircraft reaches the target
  - The point of closest approach is the minimum slant range in the cross-track plane
  - This requirement allows time for a night-vision camera with cross-track pointing capability to be pointed at the target as the aircraft flies overhead (or passes by)
  - The **goal** is to increase warning time up to **60 sec**
    - Bonus evaluation points will be assigned for this enhanced capability

- **Doppler measurement:** target Doppler measurement may be simultaneous (parallel processing) with target detection or sequential radar mode following target detection. Doppler measurement may use all available pulses for multiple scans (fore & aft) to meet the Doppler precision spec.

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- Doppler Measurement Precision
  - (RMS Error = δf_d): ≤ 700 Hz; goal: ≤ 50 Hz
- For purposes of evaluating the rms error, the following equation shall be used:

$$\delta f_d = \frac{1}{(TOT) * \sqrt{2E_b/N_o}}$$

Note T/N ratio – clutter does not affect signal-to-noise ratio

  - where $2E/N_o$ is the max output from a matched filter receiver
  - E = echo energy/pulse = P_r * tau (<u>single pulse</u>)
  - $N_o = kT_{sys}$
  - TOT = time on target (additive for multiple scans)
    = #scans*(N-1)/PRF where N = POT/scan

- **Doppler Measurement Analysis:** Assume that the aircraft's ground velocity (relative to non-moving terrain) is known without error, which allows the clutter Doppler to be precisely calculated at any point in the antenna footprint and azimuth scan position

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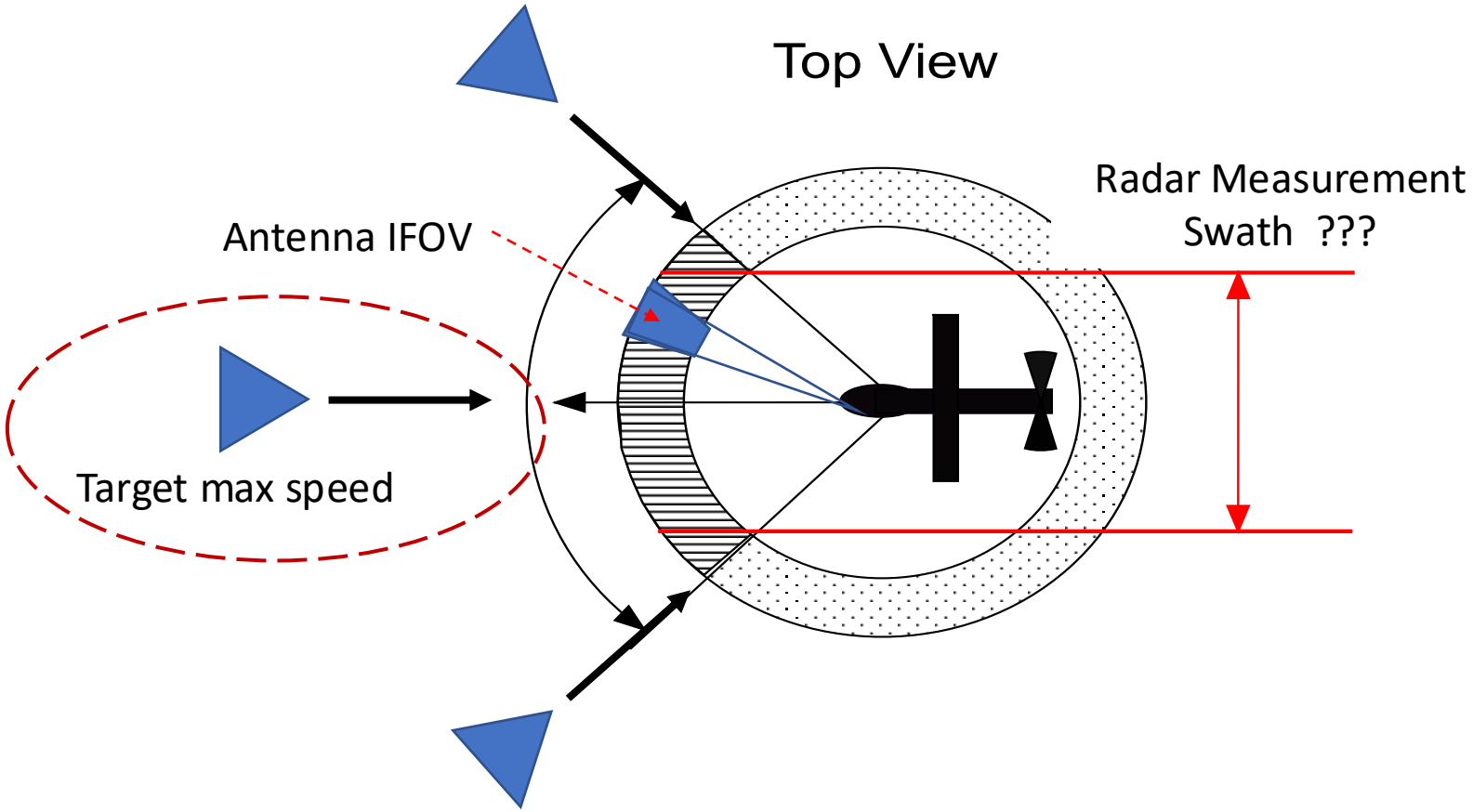
- Perform a systems analysis to verify that you can measure the target Doppler in the presence of terrain clutter Doppler (i.e., that they are separable in the freq domain)

- **Doppler Measurement Analysis:** Assume that the aircraft's ground velocity (relative to non-moving terrain) is known without error, which allows the clutter Doppler to be precisely calculated at any point in the antenna footprint and azimuth scan position
- Perform a systems analysis to verify that you can measure the target Doppler in the presence of terrain clutter Doppler (i.e., that they are separable in the freq domain)
- This requirement must be verified by analysis using a worst-case scenario for the target speed and direction relative to the radar look direction. <u>This analysis shall be presented in your proposal assuming the following worst-case conditions:</u>

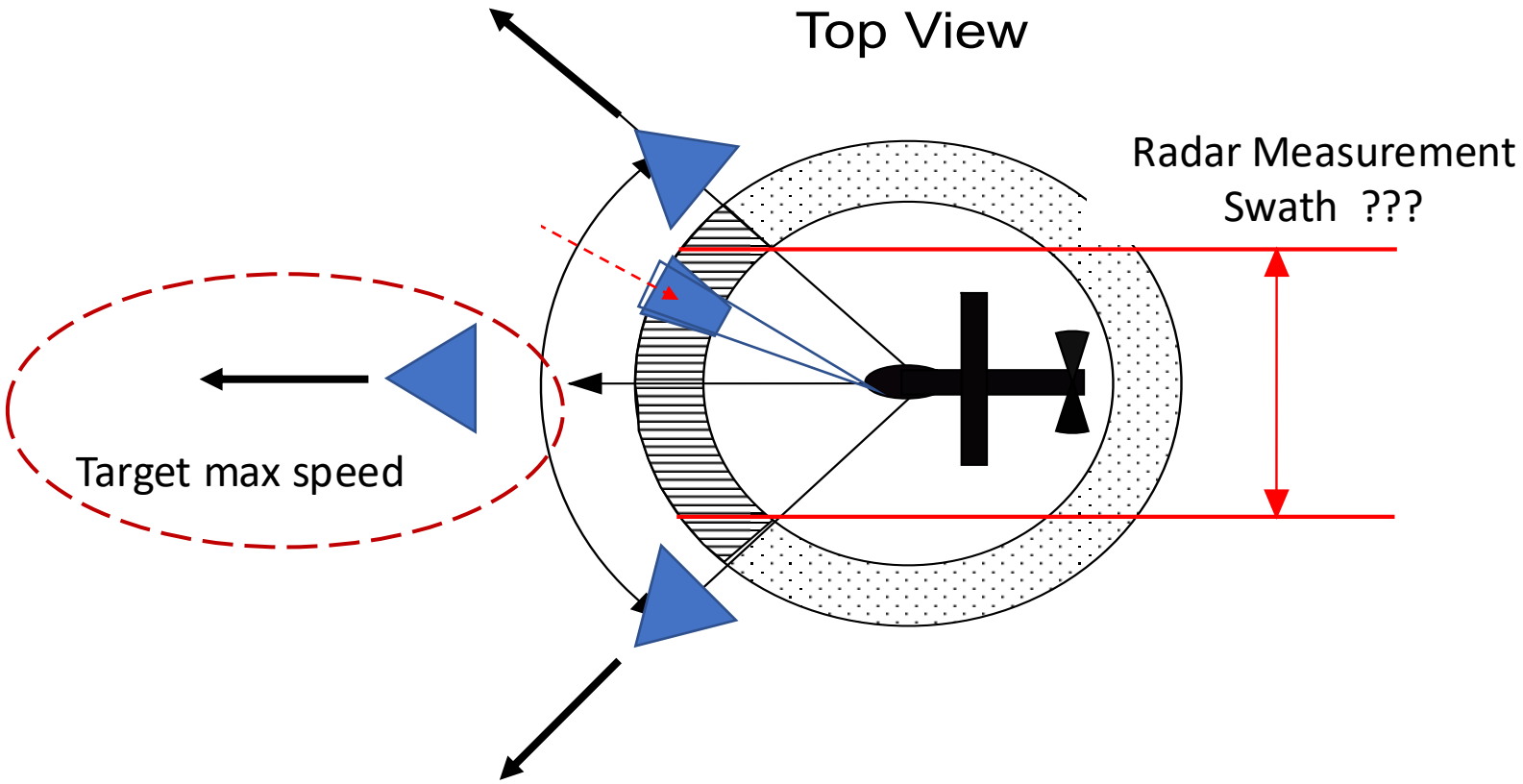
This reqmt must be verified by analysis using a worst-case scenario for the target speed and direction relative to the radar look direction. <u>This analysis shall be presented in your proposal assuming the following worst-case conditions:</u>

(1) max target speed and relative azimuth of 0° (3 Az directions: along-track and @ edges of active scan) with respect to the radar look direction (approaching the radar);



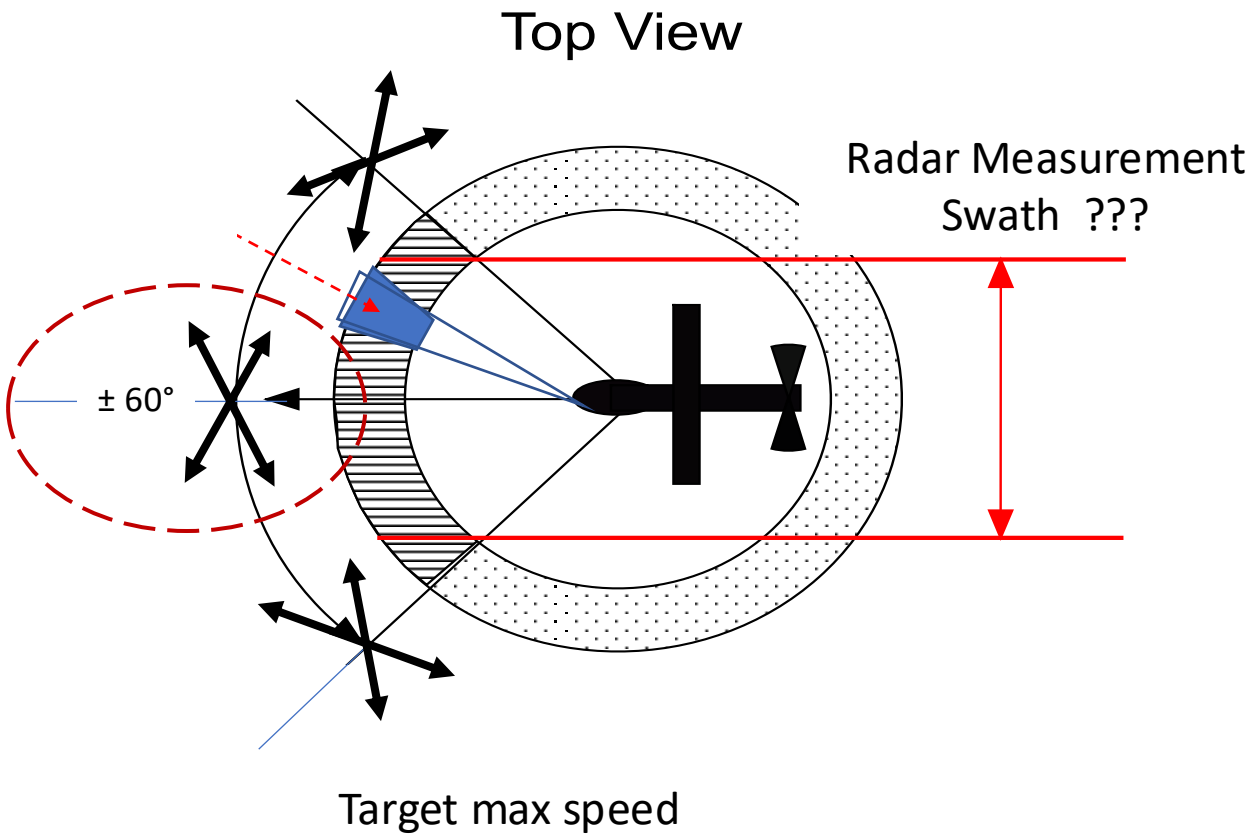
This reqmt must be verified by analysis using a worst-case scenario for the target speed and direction relative to the radar look direction. <u>This analysis shall be presented in your proposal assuming the following worst-case conditions:</u>

(2) max target speed and relative azimuth of 180° with respect to the radar look direction (moving directly away from radar);



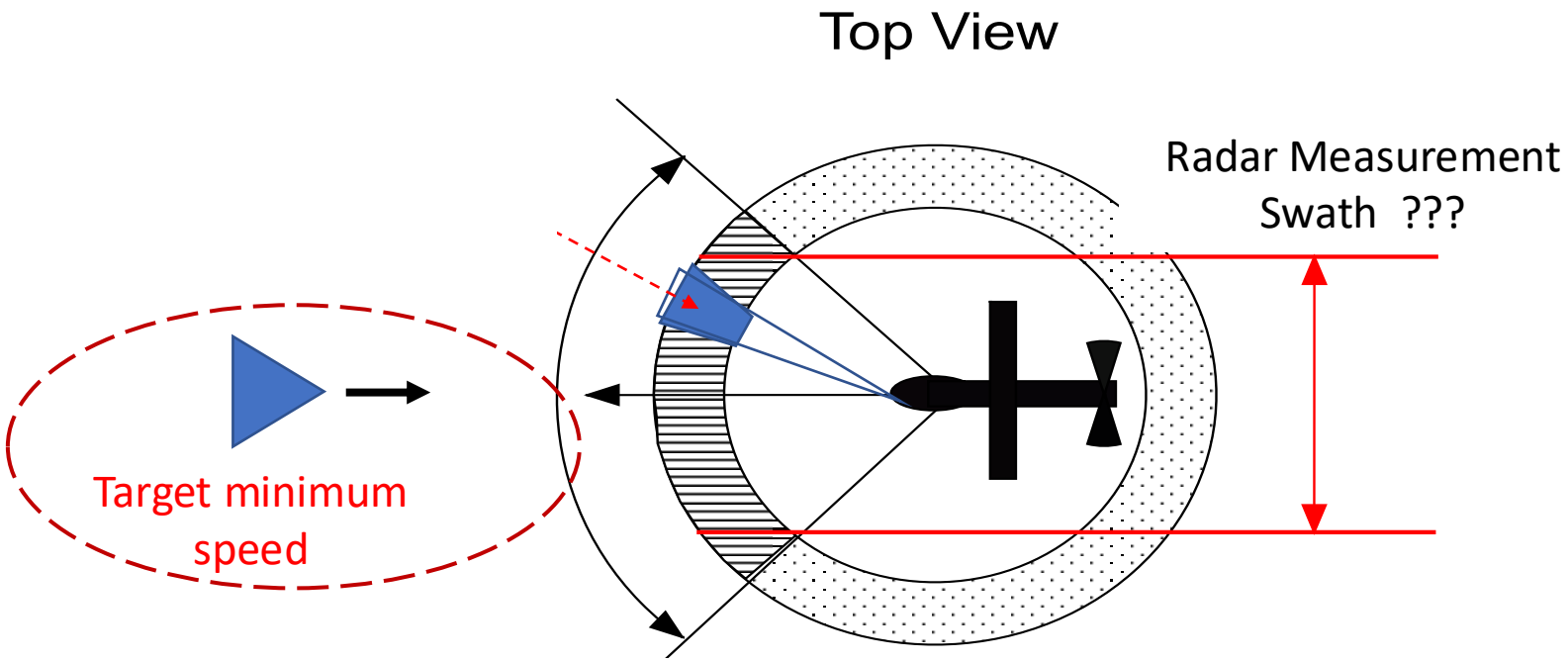
This reqmt must be verified by analysis using a worst-case scenario for the target speed and direction relative to the radar look direction. <u>This analysis shall be presented in your proposal assuming the following worst-case conditions:</u>

(3) max target speed and relative azimuth of ± 60° with respect to the radar look direction (3 Az directions: along-track and @ edges of active scan) ;



This reqmt must be verified by analysis using a worst-case scenario for the target speed and direction relative to the radar look direction. <u>This analysis shall be presented in your proposal assuming the following worst-case conditions:</u>

(4) Minimum target speed and relative azimuth of 0° with respect to the radar look direction (3 Az directions: along-track and @ edges of active scan);



- **Target characteristics:** Assume that the target will travel in a straight-line trajectory in any direction with a maximum speed of 170 km/hr (47.3 m/s)

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- Assume that the target major dimension is 6 m, which requires that the pulse resolution ≥ 6 m for both range and azimuth to assure that the target occupies only a single range bin

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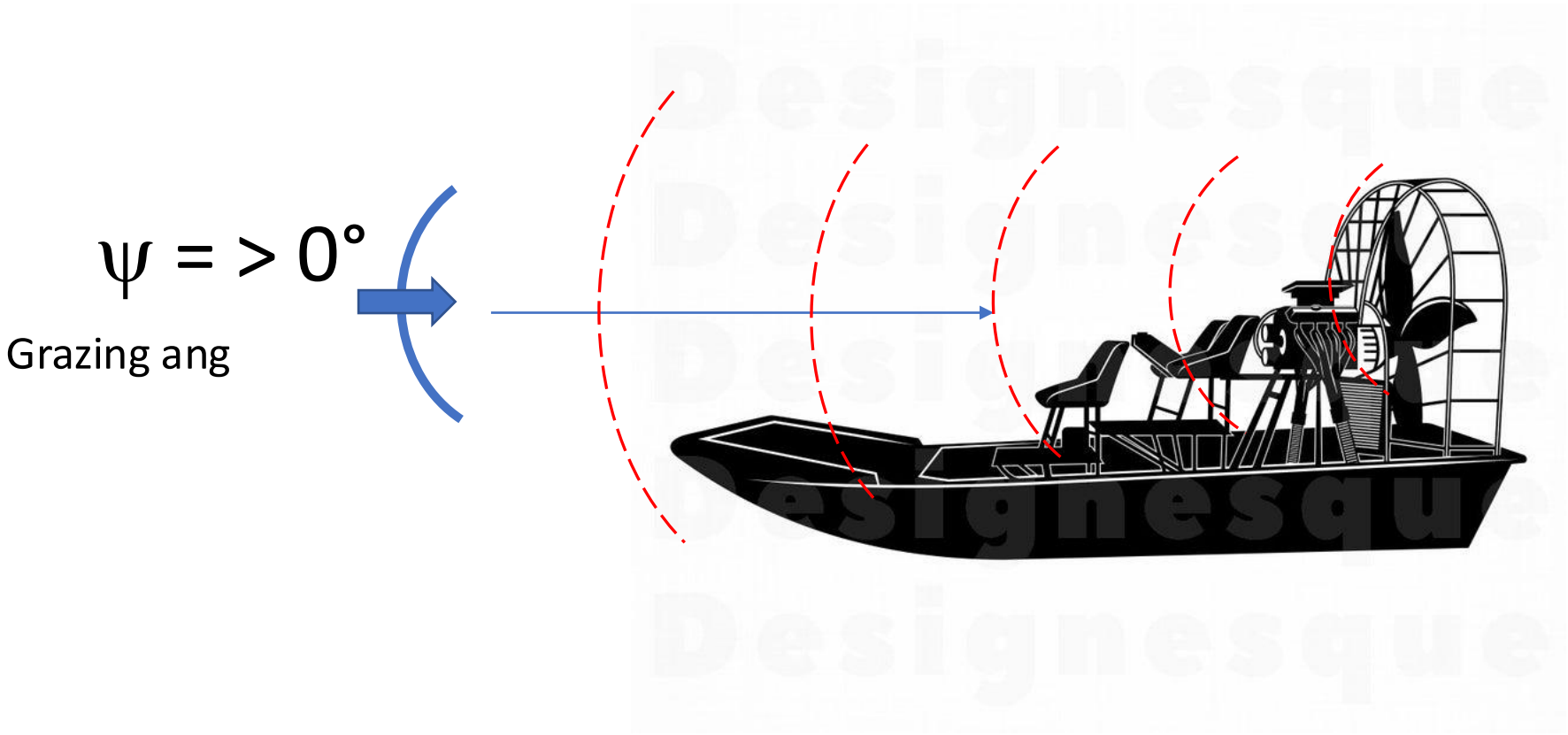
- In any antenna scan, assume that only a single target (air boat) is present within the radar measurement swath (R_{near} to R_{far} , ± ??? az)
- Assume that the target major dimension is 6 m, which requires that the pulse resolution ≥ 6 m for both range and azimuth to assure that the target occupies only a single range bin
- Also, for the radar analysis, assume that, when the antenna sweeps across the target, the target will be in the center of a range bin (i.e., the target will not split into two range bins)

Radar Cross Section (RCS)
(Independent of Azimuth)



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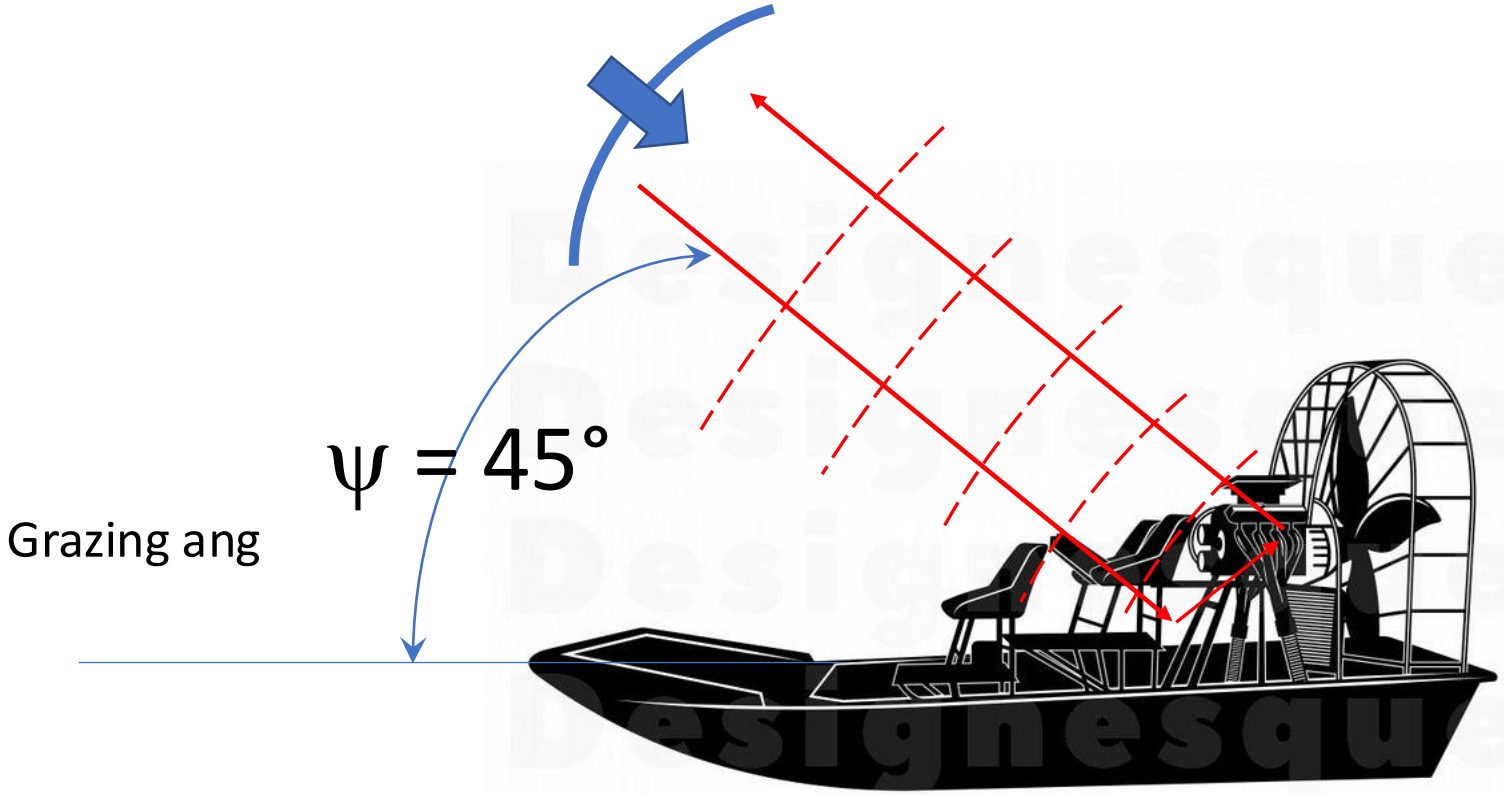

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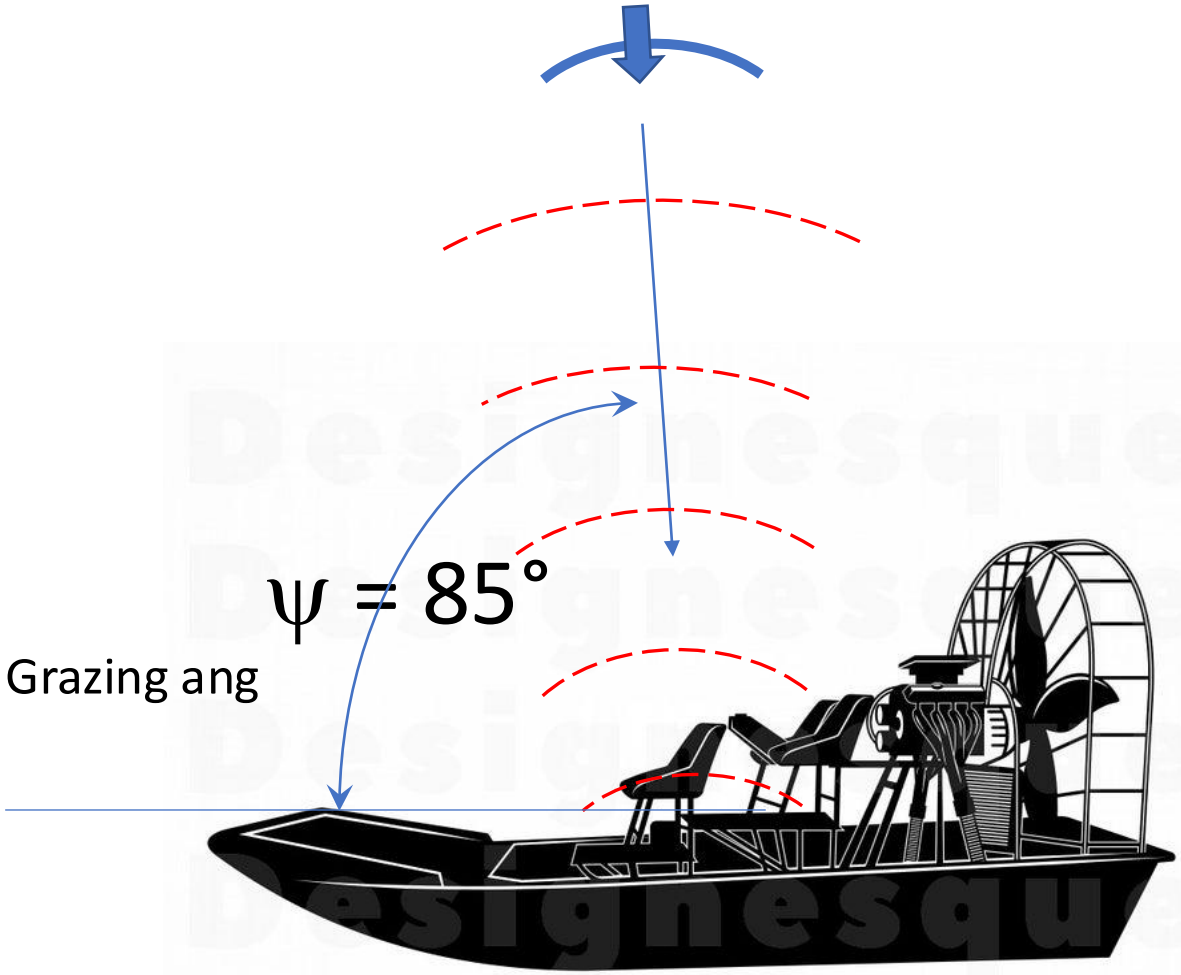
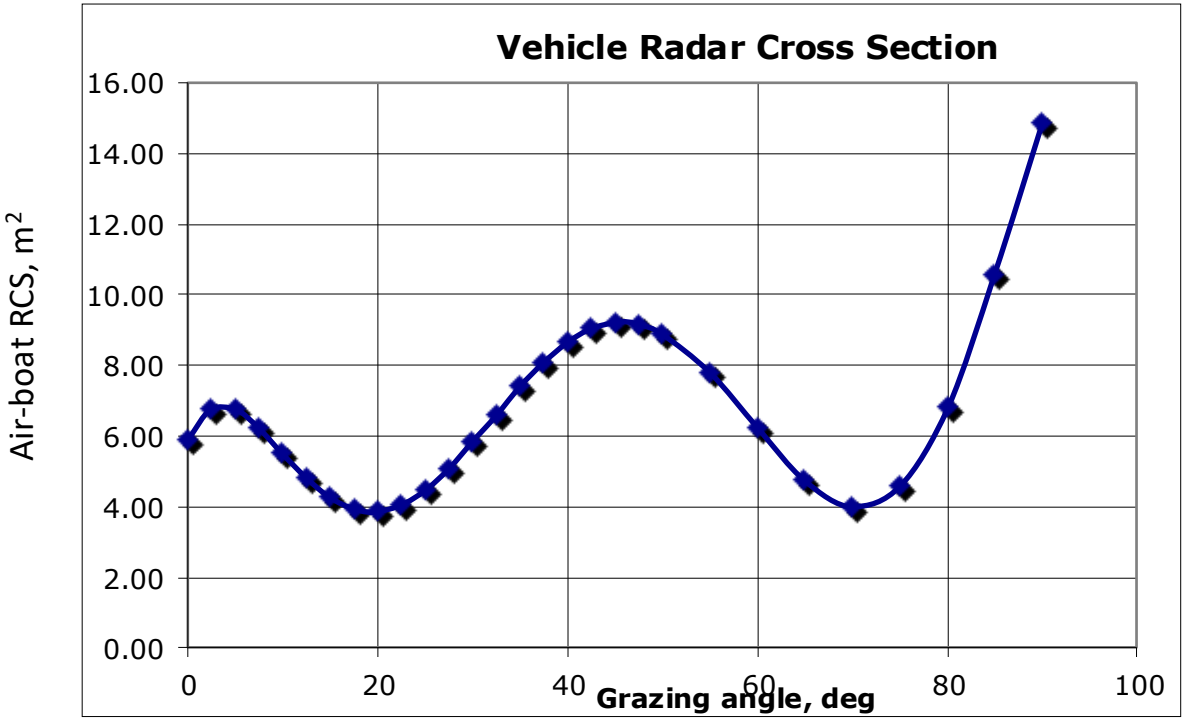


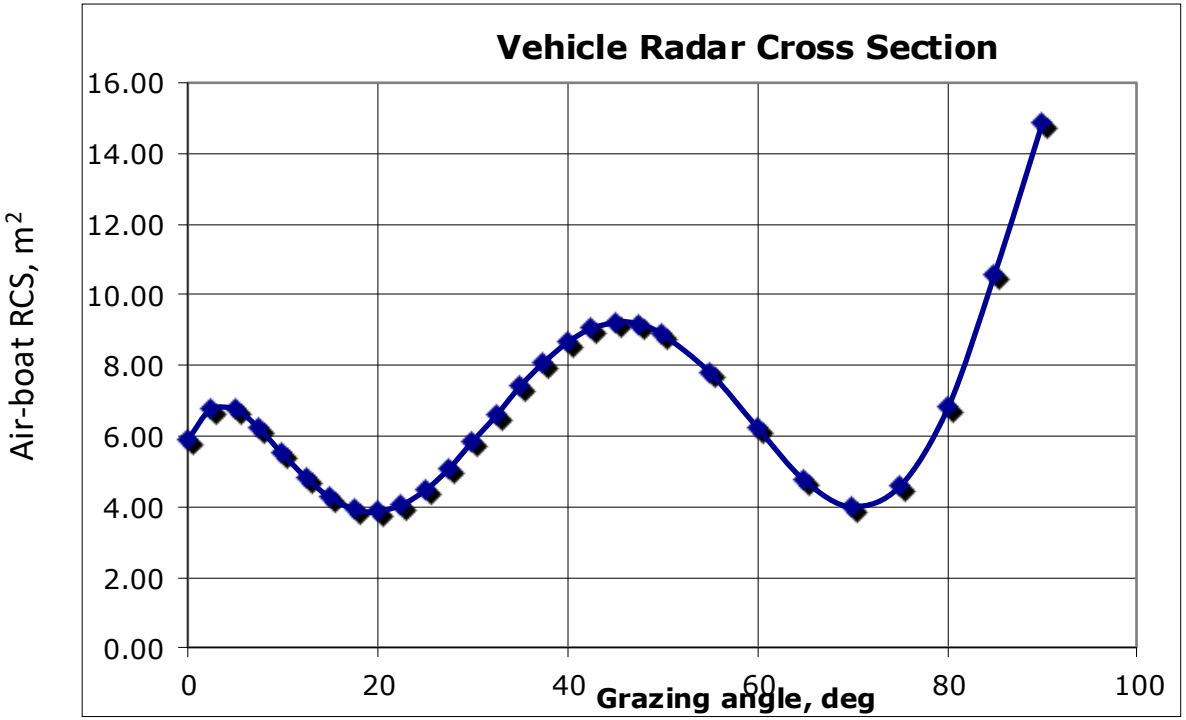
- Target radar cross section changes as a function of grazing angle

Note: this curve is a “text case” for your model



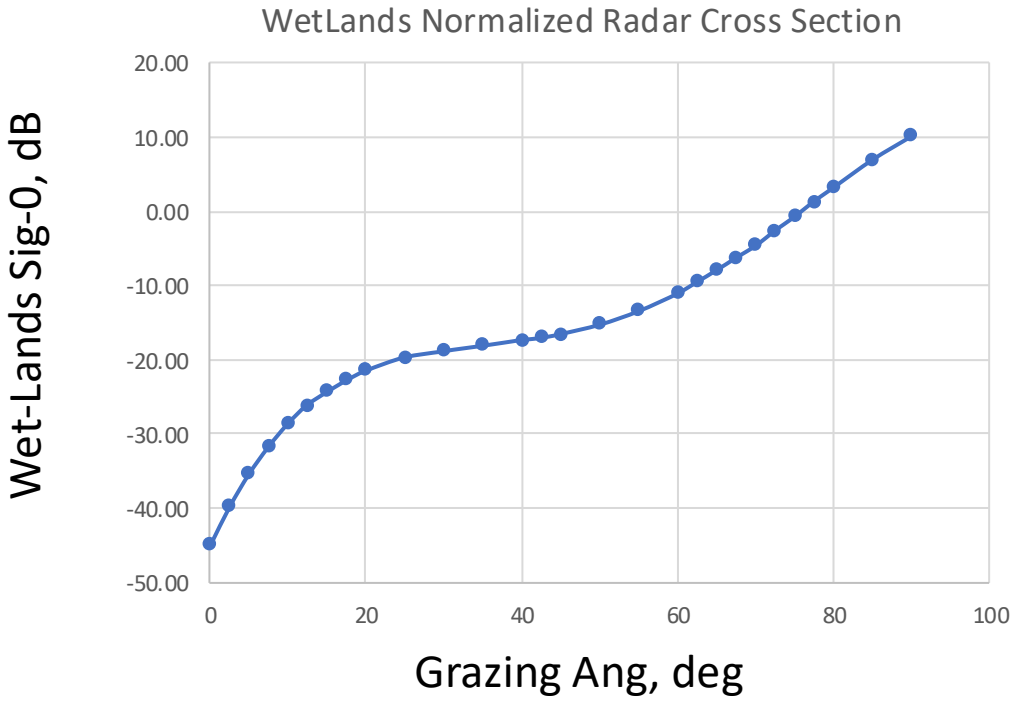
- Target radar cross section changes as a function of grazing angle
- The relation between sigma (square meters) and grazing angle is expressed as a 6th order polynomial in the grazing angle
- % polyfit
- % y = p1*x^6 + p2*x^5 + p3*x^4 + p4*x^3 + p5*x^2 + p6*x^ + p7
- % p1 = -4.41228E-09% (negative)
- % p2 = 1.24415E-06
- % p3 = -1.2775E-04 (negative)
- % p4 = 5.7724E-03
- % p5 = -0.1075 (negative)
- % p6 = 0.5756
- % p7 = 5.90

Note: this curve is a "text case" for your model



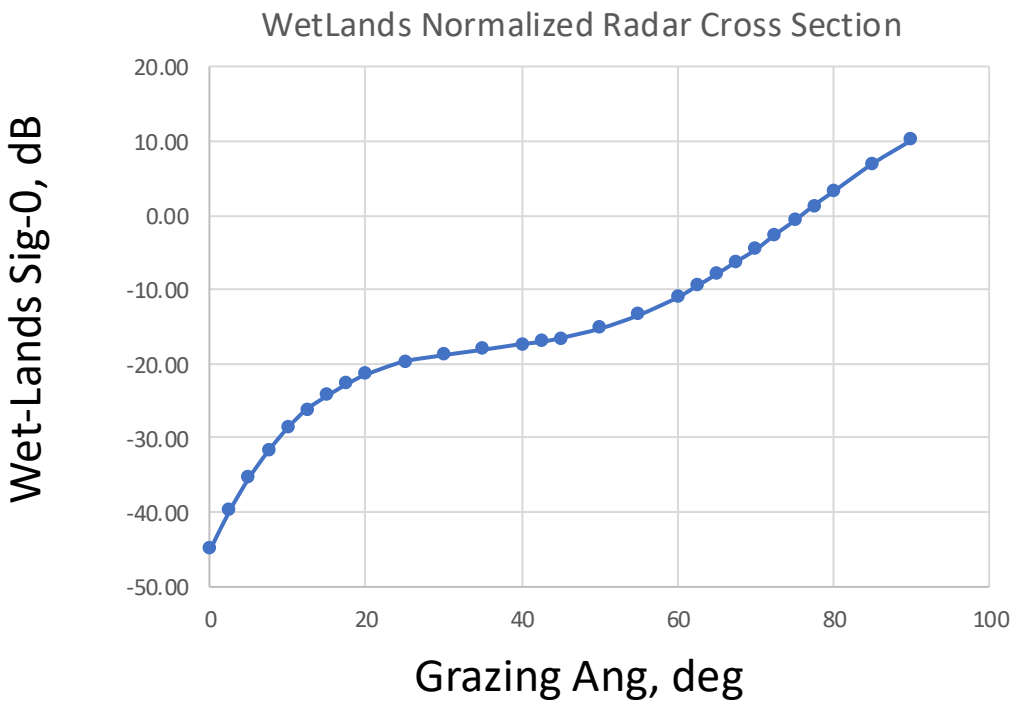
- **Clutter characteristics:** Assume that the “wetland” terrain normalized radar cross section (sigma-0) is a function of grazing angle

Note: this curve is a “text case” for your model



- **Clutter characteristics:** Assume that the “wetland” terrain normalized radar cross section (sigma-0) is a function of grazing angle
- assume that the clutter echo signal is random with fast fading characteristics like Gaussian noise
- The relation between sigma-0 (dBsq-m) and grazing angle is expressed as a 4th order polynomial in the grazing angle
- %polyfit
- % y = p1*x^4 + p2*x^3 + p3*x^2 + p4*x + p5
- % Coefficients:
- % p1 = -4.3868e-06
- % p2 = 0.00098614
- % p3 = -0.071347
- % p4 = 2.2443
- % p5 = -44.959

Note: this curve is a “text case” for your model



Preproposal Test Case Evaluation

- It is highly recommended that you submit intermediate design calculations for evaluation
- For this purpose
  - The following table must be completed
    - But, only related parameters for the given analysis are required

Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
Antenna Azimuth +/- from fore-look		deg
Xmtr Pwr		W
Xmt pulse width		microsec
PRF		Hz
<Tfa>		sec
Design capability: # scans on target = N		integer
PCR		
Detection to Doppler (Time/scan)*		ratio

Freq must always be provided, but others are optional for a given analysis

N ≥ 3

PCR ≥ 1.0

Preproposal Test Case Evaluation

- It is highly recommended that you submit intermediate design calculations for evaluation
- For this purpose
  - The following table must be completed
    - But, only related parameters for the given analysis are required
- However, for the final proposal all parameters must be provided
  - Failure to supply the required information is basis for being declared as **NON-COMPLIANT**!

Radar Design Parameters Choices		
Radar Freq		Hz
UAV Altitude		km
Ant.boresight grazing ang		deg
Ant dia- elevation		meter
Ant dia- azimuth		meter
Ant spin rate		rpm
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Xmtr Pwr		W
Xmt pulse width		microsec
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Design capability: # scans on target = N		integer
PCR		
Detection to Doppler (Time/scan)*		ratio

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N ≥ 3

PCR ≥ 1.0

Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?

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Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?
2. UAV altitude: _____ (km) ?
3. Antenna boresight grazing angle: _____ (deg)?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?
2. UAV altitude: _____ (km) ?
3. Antenna boresight grazing angle: _____ (deg)?
4. Antenna elevation aperture dimension: _____ (m)?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?
2. UAV altitude: _____ (km) ?
3. Antenna boresight grazing angle: _____ (deg)?
4. Antenna elevation aperture dimension: _____ (m)?
5. Antenna azimuth aperture dimension: _____ (m)?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?
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5. Antenna azimuth aperture dimension: _____ (m)?
6. Antenna spin rate: _____ (rpm)?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

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6. Antenna spin rate: _____ (rpm)?
7. Antenna azimuth limit (± deg for active-radar-meas): _____ (deg)?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

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2. UAV altitude: _____ (km) ?
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7. Antenna azimuth limit (± deg for active-radar-meas): _____ (deg)?
8. Transmitter power: _____ (W)?

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6. Antenna spin rate: _____ (rpm)?
7. Antenna azimuth limit (± deg for active-radar-meas): _____ (deg)?
8. Transmitter power: _____ (W)?
9. Xmt pulse width: _____ (microsec)?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?
2. UAV altitude: _____ (km) ?
3. Antenna boresight grazing angle: _____ (deg)?
4. Antenna elevation aperture dimension: _____ (m)?
5. Antenna azimuth aperture dimension: _____ (m)?
6. Antenna spin rate: _____ (rpm)?
7. Antenna azimuth limit (± deg for active-radar-meas): _____ (deg)?
8. Transmitter power: _____ (W)?
9. Xmt pulse width: _____ (microsec)?
10. PRF: _____ (Hz) ?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?
2. UAV altitude: _____ (km) ?
3. Antenna boresight grazing angle: _____ (deg)?
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6. Antenna spin rate: _____ (rpm)?
7. Antenna azimuth limit (± deg for active-radar-meas): _____ (deg)?
8. Transmitter power: _____ (W)?
9. Xmt pulse width: _____ (microsec)?
10. PRF: _____ (Hz) ?
11. Average time between false alarms <Tfa>: _____ (sec)?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?
2. UAV altitude: _____ (km) ?
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6. Antenna spin rate: _____ (rpm)?
7. Antenna azimuth limit (± deg for active-radar-meas): _____ (deg)?
8. Transmitter power: _____ (W)?
9. Xmt pulse width: _____ (microsec)?
10. PRF: _____ (Hz) ?
11. Average time between false alarms <Tfa>: _____ (sec)?
12. Number of fore-looking scans that target can be tracked _____ ?

Radar Design Parameters (Simultaneous Target Detection & Doppler):

1. Radar Freq: _____ (Hz) ?
2. UAV altitude: _____ (km) ?
3. Antenna boresight grazing angle: _____ (deg)?
4. Antenna elevation aperture dimension: _____ (m)?
5. Antenna azimuth aperture dimension: _____ (m)?
6. Antenna spin rate: _____ (rpm)?
7. Antenna azimuth limit (± deg for active-radar-meas): _____ (deg)?
8. Transmitter power: _____ (W)?
9. Xmt pulse width: _____ (microsec)?
10. PRF: _____ (Hz) ?
11. Average time between false alarms <Tfa>: _____ (sec)?
12. Number of fore-looking scans that target can be tracked _____ ?
13. Pulse compression ratio: _____ (PCR)?

Radar Design Parameters (Sequential Target Detection & Doppler):

In addition to previous page, provide:

1. Fwd-look POT/scan used for Target detection: _____ (#) ?

Radar Design Parameters (Sequential Target Detection & Doppler):

In addition to previous page, provide:

1. Fwd-look POT/scan used for Target detection: _____ (#) ?
2. Target worst-case cum P_d(2 of 3) = _____ and location (range-far or near) _____ ?

Radar Design Parameters (Sequential Target Detection & Doppler):

In addition to previous page, provide:

1. Fwd-look POT/scan used for Target detection: _____ (#) ?
2. Target worst-case cum P_d(2 of N) = _____ and location (range-far or near) _____ ?
3. PRF Doppler mode: _____ (Hz) ?

Radar Design Parameters (Sequential Target Detection & Doppler):

In addition to previous page, provide:

1. Fwd-look POT/scan used for Target detection: _____ (#) ?
2. Target worst-case cum P_d(2 of N) = _____ and location (range-far or near) _____ ?
3. PRF Doppler mode: _____ (Hz) ?
4. Xmtr duty cycle Doppler mode: _____ microsec?

Radar Design Parameters (Sequential Target Detection & Doppler):

In addition to previous page, provide:

1. Fwd-look POT/scan used for Target detection: _____ (#) ?
2. Target worst-case cum P_d(2 of N) = _____ and location (range-far or near) _____ ?
3. PRF Doppler mode: _____ (Hz) ?
4. Xmtr duty cycle Doppler mode: _____ microsec?
5. (Target detection time/scan)/(TOT/scan): _____ (ratio, req'd for sequential Doppler meas)?

Other Required Proposal Material

As a minimum, the proposal shall include the following:

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

- Minimum grazing angle (2-way pattern): _____ deg?

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

- Minimum grazing angle (2-way pattern): _____ deg?
- Range @ near, bore and far? R_{near}= _____ m ; R_{bore}= _____ m; and R_{far}= _____ m?

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

- Minimum grazing angle (2-way pattern): _____ deg?
- Range @ near, bore and far? R_{near}= _____ m ; R_{bore}= _____ m; and R_{far}= _____ m?
- Min Ant spin-rate _____ rpm to assure no missed target for specified _____# scans

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

- Minimum grazing angle (2-way pattern): _____ deg?
- Range @ near, bore and far? R_{near}= _____ m ; R_{bore}= _____ m; and R_{far}= _____ m?
- Min Ant spin-rate _____ rpm to assure no missed target for specified _____# scans
- Active radar measurements Az scan limit (± Az relative to fore-look): _____ deg

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

- Minimum grazing angle (2-way pattern): _____ deg?
- Range @ near, bore and far? R_{near}= _____ m ; R_{bore}= _____ m; and R_{far}= _____ m?
- Min Ant spin-rate _____ rpm to assure no missed target for specified _____# scans
- Active radar measurements Az scan limit (± Az relative to fore-look): _____ deg
- Minimum warning time (@ worst case target location): _____ (sec)?

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

- Minimum grazing angle (2-way pattern): _____ deg?
- Range @ near, bore and far? R_{near}= _____ m ; R_{bore}= _____ m; and R_{far}= _____ m?
- Min Ant spin-rate _____ rpm to assure no missed target for specified _____# scans
- Active radar measurements Az scan limit (± Az relative to fore-look): _____ deg
- Minimum warning time (@ worst case target location): _____ (sec)?
- Radar measurement swath width: _____ km?

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

- Minimum grazing angle (2-way pattern): _____ deg?
- Range @ near, bore and far? R_{near}= _____ m ; R_{bore}= _____ m; and R_{far}= _____ m?
- Min Ant spin-rate _____ rpm to assure no missed target for specified _____# scans
- Active radar measurements Az scan limit (± Az relative to fore-look): _____ deg
- Minimum warning time (@ worst case target location): _____ (sec)?
- Radar measurement swath width: _____ km?
- Pulse surface footprint @ boresight, range resolution: _____ m?

  azimuth resolution: _____ m?

2-way ant pattern

Radar Performance Achieved:

Consecutive (Target detection & Doppler) or Sequential (Target detection & Doppler)?

- Provide radar mode timing diagram for single scan for sequential operation

- Minimum grazing angle (2-way pattern): _____ deg?
- Range @ near, bore and far? R_{near}= _____ m ; R_{bore}= _____ m; and R_{far}= _____ m?
- Min Ant spin-rate _____ rpm to assure no missed target for specified _____# scans
- Active radar measurements Az scan limit (± Az relative to fore-look): _____ deg
- Minimum warning time (@ worst case target location): _____ (sec)?
- Radar measurement swath width: _____ km?
- Pulse surface footprint @ boresight, range resolution: _____ m?

  azimuth resolution: _____ m?

2-way ant pattern

- Transmitter duty cycle: _____ during target detection?

  _____ during Doppler measurement?

Radar Performance Achieved cont.

- Target radar cross section @ boresight: _____ m^2 ?

Radar Performance Achieved cont.

- Target radar cross section @ boresight: _____ m^2 ?
- Terrain radar cross section (“clutter’) @ boresight: _____ m^2 ?

Radar Performance Achieved cont.

- Target radar cross section @ boresight: _____ m^2 ?
- Terrain radar cross section (“clutter’) @ boresight: _____ m^2 ?
- Total # pulse/scan (POT) used for target detection: _____ # ?

Radar Performance Achieved cont.

- Target radar cross section @ boresight: _____ m^2 ?
- Terrain radar cross section ("clutter') @ boresight: _____ m^2 ?
- Total # pulse/scan (POT) used for target detection: _____ # ?
- Given radar equation: P_r = X * (sigma / R^4), X @ near-range; = _____ dB?

= _____ @ boresight; & = _____ @ far-range ?

Radar Performance Achieved cont.

- Target radar cross section @ boresight: _____ m^2 ?
- Terrain radar cross section ("clutter') @ boresight: _____ m^2 ?
- Total # pulse/scan (POT) used for target detection: _____ # ?
- Given radar equation: P_r = X * (sigma / R^4), X @ near-range; = _____ dB?

= _____ @ boresight; & = _____ @ far-range ?

- The T/(C+N) in dB:

= _____ @ near-range; = _____ @ boresight; & = _____ @ far-range ?

Radar Performance Achieved cont.

- Target radar cross section @ boresight: _____ m^2 ?
- Terrain radar cross section ("clutter') @ boresight: _____ m^2 ?
- Total # pulse/scan (POT) used for target detection: _____ # ?
- Given radar equation: P_r = X * (sigma / R^4), X @ near-range; = _____ dB?

= _____ @ boresight; & = _____ @ far-range ?

- The T/(C+N) in dB:

= _____ @ near-range; = _____ @ boresight; & = _____ @ far-range ?

- Multi-scan P_d(2 of 3 scans):

= _____ @ near-range; = _____ @ boresight; & = _____ @ far-range ?

Radar Performance Achieved cont.

- Target radar cross section @ boresight: _____ m^2 ?
- Terrain radar cross section ("clutter') @ boresight: _____ m^2 ?
- Total # pulse/scan (POT) used for target detection: _____ # ?
- Given radar equation: P_r = X * (sigma / R^4), X @ near-range; = _____ dB?

= _____ @ boresight; & = _____ @ far-range ?

- The T/(C+N) in dB:

= _____ @ near-range; = _____ @ boresight; & = _____ @ far-range ?

- Multi-scan P_d(2 of 3 scans):

= _____ @ near-range; = _____ @ boresight; & = _____ @ far-range ?

- Max Doppler freq @ boresight: _____ Hz
- Doppler TOT for selected # scans: _____ sec
- Doppler measurement precision @ boresight: _____ Hz

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Transmitted power, W	≤ 1500	(xxx)	(1500 – xxx)
• Goal, W	≤ 10		

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Transmitted power, W	≤ 1500	(xxx)	(1500 – xxx)
• Goal, W	≤ 10		
Transmitter duty cycle	≤ 0.10	(xxx)	(0.10 – xxx)
(Doppler Meas)	≤ 0.20	(xxx)	(0.20 – xxx)

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Transmitted power, W	≤ 1500	(xxx)	(1500 – xxx)
• Goal, W	≤ 10		
Transmitter duty cycle	≤ 0.10	(xxx)	(0.10 – xxx)
(Doppler Meas)	≤ 0.20	(xxx)	(0.20 – xxx)
Antenna Grazing angle @ R_{far}, deg	≥ 2.0	(xxx)	(xxx - 2)

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Transmitted power, W	≤ 1500	(xxx)	(1500 – xxx)
• Goal, W	≤ 10		
Transmitter duty cycle	≤ 0.10	(xxx)	(0.10 – xxx)
(Doppler Meas)	≤ 0.20	(xxx)	(0.20 – xxx)
Antenna Grazing angle @ R_{far}, deg	≥ 2.0	(xxx)	(xxx - 2)
Antenna Az dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Transmitted power, W	≤ 1500	(xxx)	(1500 – xxx)
• Goal, W	≤ 10		
Transmitter duty cycle	≤ 0.10	(xxx)	(0.10 – xxx)
(Doppler Meas)	≤ 0.20	(xxx)	(0.20 – xxx)
Antenna Grazing angle @ R_{far}, deg	≥ 2.0	(xxx)	(xxx - 2)
Antenna Az dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		
Antenna El dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Transmitted power, W	≤ 1500	(xxx)	(1500 – xxx)
• Goal, W	≤ 10		
Transmitter duty cycle	≤ 0.10	(xxx)	(0.10 – xxx)
(Doppler Meas)	≤ 0.20	(xxx)	(0.20 – xxx)
Antenna Grazing angle @ R_{far}, deg	≥ 2.0	(xxx)	(xxx - 2)
Antenna Az dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		
Antenna El dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		
Measurement Swath, km	≥ 10	(xxx)	(xxx - 10)
• GOAL, km	50		

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Transmitted power, W	≤ 1500	(xxx)	(1500 – xxx)
• Goal, W	≤ 10		
Transmitter duty cycle	≤ 0.10	(xxx)	(0.10 – xxx)
(Doppler Meas)	≤ 0.20	(xxx)	(0.20 – xxx)
Antenna Grazing angle @ R_{far}, deg	≥ 2.0	(xxx)	(xxx - 2)
Antenna Az dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		
Antenna El dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		
Measurement Swath, km	≥ 10	(xxx)	(xxx - 10)
• GOAL, km	50		
Antenna spin rate	≤ 180 rpm	(xxx)	(180 – xxx)
• Goal, rpm	≤ 30		

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Transmitted power, W	≤ 1500	(xxx)	(1500 – xxx)
• Goal, W	≤ 10		
Transmitter duty cycle	≤ 0.10	(xxx)	(0.10 – xxx)
(Doppler Meas)	≤ 0.20	(xxx)	(0.20 – xxx)
Antenna Grazing angle @ R_{far}, deg	≥ 2.0	(xxx)	(xxx - 2)
Antenna Az dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		
Antenna El dimension, m	≤ 1.5	(xxx)	(1.2 – xxx)
• Goal, m	0.10		
Measurement Swath, km	≥ 10	(xxx)	(xxx - 10)
• GOAL, km	50		
Antenna spin rate	≤ 180 rpm	(xxx)	(180 – xxx)
• Goal, rpm	≤ 30		
Target tracking min # scans	≥ 3	(xxx)	(xxx - 3)
• Goal	10		

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Max PRF, Hz	≤ 500,000	(xxx)	(5E+05 – xxx)

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Max PRF, Hz	≤ 500,000	(xxx)	(5E+05 – xxx)
Pulse IFOV on surface			
del-Range, m	≥ 6.0	(xxx)	(xxx - 6)
del-Az, m	≥ 6.0	(xxx)	(xxx - 6)

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Max PRF, Hz	≤ 500,000	(xxx)	(5E+05 – xxx)
Pulse IFOV on surface			
del-Range, m	≥ 6.0	(xxx)	(xxx - 6)
del-Az, m	≥ 6.0	(xxx)	(xxx - 6)
$<T_{fa}>$ (derived), sec	$\geq 10*T_{spin}$	(xxx)	(xxx – 10*Ts)

T_{spin} = 60/(ant rpm) , sec

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Max PRF, Hz	≤ 500,000	(xxx)	(5E+05 – xxx)
Pulse IFOV on surface			
del-Range, m	≥ 6.0	(xxx)	(xxx - 6)
del-Az, m	≥ 6.0	(xxx)	(xxx - 6)
$<T_{fa}>$ (derived), sec	$\geq 10*T_{spin}$	(xxx)	(xxx – 10*Ts)
T_{spin} = 60/(ant rpm) , sec			
Target detection capability (worst-case)			
P_d(2 of "N")	≥ 0.80	(xxx)	(xxx – 0.80)

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Max PRF, Hz	≤ 500,000	(xxx)	(5E+05 – xxx)
Pulse IFOV on surface			
del-Range, m	≥ 6.0	(xxx)	(xxx - 6)
del-Az, m	≥ 6.0	(xxx)	(xxx - 6)
$<T_{fa}>$ (derived), sec	$\geq 10*T_{spin}$	(xxx)	(xxx – 10*Ts)
T_{spin} = 60/(ant rpm) , sec			
Target detection capability (worst-case)			
P_d(2 of "N")	≥ 0.80	(xxx)	(xxx – 0.80)
Target warning time, sec	≥ 6.0	(xxx)	(xxx - 6)
Goal, sec	≥ 60		

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Max PRF, Hz	≤ 500,000	(xxx)	(5E+05 – xxx)
Pulse IFOV on surface			
del-Range, m	≥ 6.0	(xxx)	(xxx - 6)
del-Az, m	≥ 6.0	(xxx)	(xxx - 6)
$<T_{fa}>$ (derived), sec	$\geq 10*T_{spin}$	(xxx)	(xxx – 10*Ts)
T_{spin} = 60/(ant rpm) , sec			
Target detection capability (worst-case)			
P_d(2 of "N")	≥ 0.80	(xxx)	(xxx – 0.80)
Target warning time, sec	≥ 6.0	(xxx)	(xxx - 6)
Goal, sec	≥ 60		
Radar Swath width, km	10	(xxx)	(xxx – 10)
Goal, km	50		

Specification Compliance Matrix (Final) * Positive margin required

Radar Capability	**Spec**	**Proposed**	**Margin***
Max PRF, Hz	≤ 500,000	(xxx)	(5E+05 – xxx)
Pulse IFOV on surface			
del-Range, m	≥ 6.0	(xxx)	(xxx - 6)
del-Az, m	≥ 6.0	(xxx)	(xxx - 6)
$<T_{fa}>$ (derived), sec	$\geq 10*T_{spin}$	(xxx)	(xxx – 10*Ts)
T_{spin} = 60/(ant rpm) , sec			
Target detection capability (worst-case)			
P_d(2 of "N")	≥ 0.80	(xxx)	(xxx – 0.80)
Target warning time, sec	≥ 6.0	(xxx)	(xxx - 6)
Goal, sec	≥ 60		
Radar Swath width, km	10	(xxx)	(xxx – 10)
Goal, km	50		
Doppler measurement precision	≤ 700 Hz	(xxx)	(700 - xxx)
Goal	50 Hz		

Good Luck!

- Don’t forget to take advantage of early checking _ along the way!

Next time Design Trade Studies